

Post Mining Land Use | Rezoning | Macquarie Coal Complex Pilot Project

Contamination Technical Assessment

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Executive summary

This Contamination Technical Assessment (CTA) has been prepared by Aurecon Australasia Pty Ltd (Aurecon) on behalf of the Department of Planning, Housing and Infrastructure (DPHI) to support the State-led rezoning of the Macquarie Coal Complex (the Precinct) and Teralba Southgate, located approximately 20 km southwest of Newcastle within the Lake Macquarie region, NSW.

The purpose of this CTA is to reinterpret existing contamination data, originally collected to support mine closure and rehabilitation, within the context of proposed rezoning and future land use scenarios identified in the Structure and Master Plan and Teralba Southgate. The assessment is scoped at the rezoning stage to inform land use compatibility and identification of developable land. It does not constitute development application (DA) level investigation or detailed remediation design.

Site Context

The Precinct comprises four former mining operations: including West Wallsend Colliery, Macquarie Coal, Preparation Plant, Westside Mine (open-cut), and Teralba Colliery (Northgate). Teralba Southgate is located approximately 2.5 kilometres south of the Precinct and is occupied by former mine infrastructure including shafts. Mining activities ceased progressively between 2001 and 2016. Current activities are limited to environmental management and rehabilitation in accordance with the Rehabilitation Management Plan (RMP 2022). The Precinct and Teralba Southgate are surrounded primarily by residential and mixed-use industrial areas.

The rezoning aims to facilitate employment-led development supported by residential, conservation, and open space land uses. A figure showing the extent of Teralba Southgate and the current Structure Plan for the Precinct is provided in Figure E1.

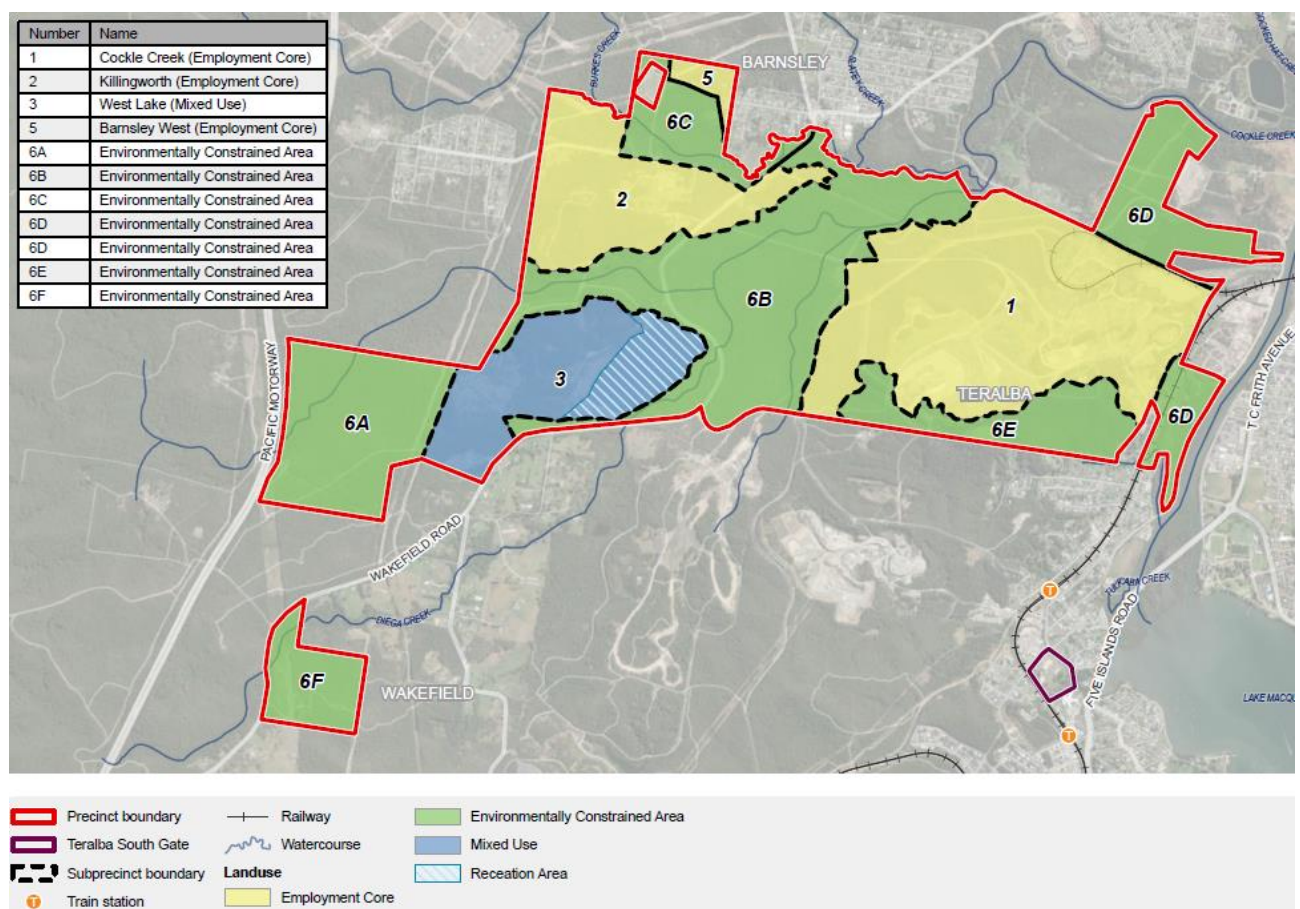


Figure E1 – Layout and extent of Teralba Southgate and the Structure Plan for the Precinct

Summary of Contamination Conditions

The Precinct encompasses five former mining operations: West Wallsend Colliery, Macquarie Coal Preparation Plant, Westside Mine (open-cut), and Teralba Colliery (Northgate). Mining operations ceased progressively across the Precinct, with Teralba Colliery closing in 2001, Westside Mine ceasing operations in 2012, and West Wallsend Colliery and Macquarie Coal Preparation Plant concluding operations in 2016. Current activities focused on long-term environmental management and rehabilitation in accordance with the Rehabilitation Management Plan (RMP 2022). The Precinct and Teralba Southgate is primarily surrounded by residential and mixed-use industrial land uses. The proposed rezoning seeks to establish employment-led development across multiple sub-precincts, with designated residential, conservation and open space.

Contamination investigations across the Precinct and Teralba Southgate have been undertaken progressively since 2005, involving:

- Approximately 400 soil and sediment sampling locations.
- 17 groundwater monitoring wells.
- Several ground gas monitoring locations.

These investigations define 16 “Domains” with variable contamination profiles. Key findings from previous investigations completed at the Precinct and Teralba Southgate include:

- Soil and sediment contamination - Elevated concentrations of heavy metals (notably zinc), total recoverable hydrocarbons (TRH), and polycyclic aromatic hydrocarbons (PAHs), primarily exceeding ecological criteria.
- Human health risks - No exceedances of commercial/industrial human health criteria identified to date.
- Asbestos - Asbestos-containing materials (ACM) confirmed in several domains.
- Acid Sulfate Soils (ASS/PASS) - Actual and potential ASS identified in specific domains at the Precinct
- Salinity - Moderate to high salinity risk across large portions of the Precinct and Teralba Southgate.
- PFAS – Principally PFOS detected above adopted ecological criteria in groundwater and surface water across multiple domains, including within the Precinct and Teralba Southgate
- Ground gas - Methane gas concentrations exceeding the lower explosive limit (LEL) identified in underground mine workings, posing a potential vapour intrusion risk, including within the Precinct (Domain 11) and Teralba Southgate (Domain 13).

A figure showing the location and extents of domains at the Precinct and Teralba Southgate is provided in Figure E2.

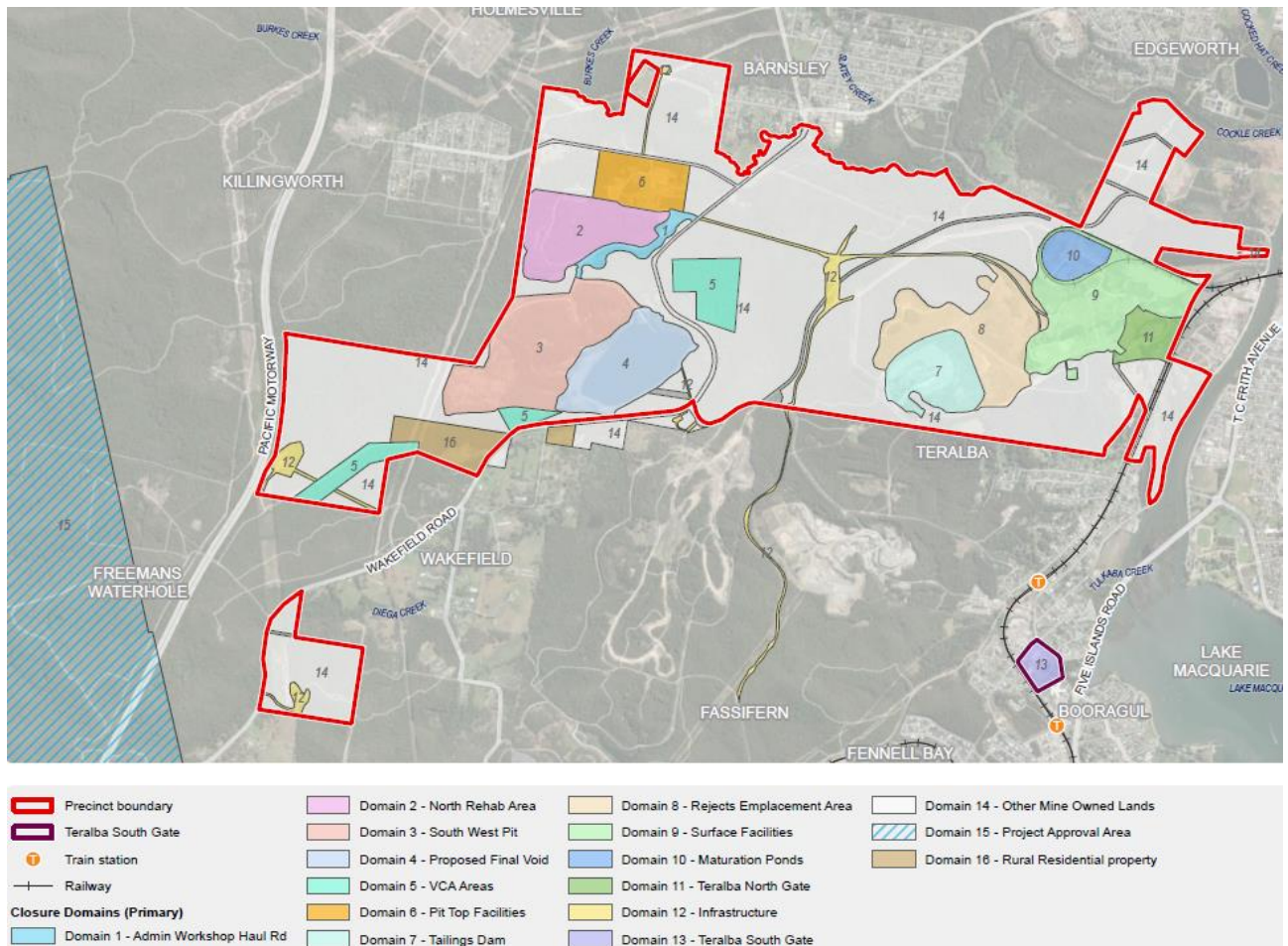


Figure E2 – Location and extent of the closure domains at the Precinct and Teralba Southgate

Data Gaps and Limitations

Significant data gaps limit ability for full characterisation of contamination and land suitability, including:

- Multiple domains with no intrusive investigation (lacking soil, groundwater, surface water and sediment data).
- Incomplete ASS and groundwater characterisation.
- Limited assessment beneath buildings and inaccessible areas/structures.
- Incomplete delineation of buried asbestos materials.
- Limited ground gas assessments.

Land Use Suitability

Based on available data and acknowledging limitations:

- Commercial and industrial uses - Generally compatible across most of the Precinct and Teralba Southgate, subject to targeted investigation and management.
- Sensitive uses (residential, conservation, recreational) - Require further investigation and risk assessment to confirm suitability.
- Key constraint - The Tailings Storage Facility (Domain 7) presents a significant constraint requiring detailed, multidisciplinary assessment.

Implementation Approach and Key Recommendations

A staged, risk-based approach to investigation and management is recommended to support rezoning and future development. With appropriate controls and management (including detailed site investigations and appropriate remediation), the Precinct and Teralba Southgate can be made suitable for its intended uses while protecting human health and environmental values, including Cockle Creek and Lake Macquarie.

Based on available data and with consideration to identified data gaps, the assessment concludes that known or potential contamination is unlikely to preclude commercial/industrial rezoning for the majority of the Precinct and Teralba Southgate, provided that targeted investigation and appropriate management controls are implemented during future design and construction phases. However, suitability determinations vary by sub-precinct and proposed land use. Employment-led commercial and industrial uses are generally considered compatible with the current contamination profile across most sub-precincts. However, more sensitive land uses including medium or high-density residential, conservation, and recreational uses, require targeted investigation and risk assessment to confirm suitability. This is particularly important at Teralba Southgate where detailed ground gas assessments have not been undertaken. The Tailings Storage Facility (Domain 7) also represents a material constraint requiring detailed assessment and integrated management across multiple environmental and engineering disciplines.

Implementation of the management framework and performance criteria outlined in this CTA, combined with progressive investigation and remediation through the DA approval process, will enable the Precinct and Teralba Southgate to support sustainable post-mining land use while protecting human health and environmental values in Cockle Creek and Lake Macquarie.

Based on the findings of this assessment, and subject to the limitations outlined in Section 11, Aurecon presents the following recommendations:

- Implement a risk-based, staged investigation program for all future development proposals, with Detailed Site Investigations (DSIs) and Tier 2/3 quantitative risk assessments undertaken where elevated COPC concentrations are present or data gaps are identified, ensuring land use suitability is confirmed prior to development approval
- Prioritise targeted investigation of Domains 2, 5, 7, 14, and 16 (uninvestigated domains), comprehensive groundwater assessment across all domains, ASS characterisation in low-lying areas, and asbestos delineation sampling to resolve material data gaps constraining sensitive land use suitability determination
- Undertake integrated assessment of the Tailings Storage Facility (TSF) across water management, geotechnical stability, dam safety, and regulatory compliance frameworks, including geochemical characterisation of tailings materials and assessment of leachate generation and Acid Mine Drainage (AMD) pathways to inform remediation and long-term management strategy
- Undertake targeted surface water and sediment quality assessments adjacent to waterway and wetland interface zones (including recreational areas within West Lake); and assess PFAS-impacted areas in accordance with the PFAS National Environmental Management Plan (NEMP 3.1)
- Undertake comprehensive ground gas assessment in accordance with NSW EPA Assessment and management of hazardous ground gases (EPA 2020), with particular focus on Domains 11 and 13 where methane concentrations were reported. If required, implement ground gas mitigation measures during building design and construction to prevent vapour intrusion risk to future enclosed structures
- Confirm land suitability for more sensitive uses (residential in Teralba Southgate, conservation, and recreational uses) through targeted investigation of soil, groundwater, surface water, and ground gas; implement PFAS plume delineation, asbestos delineation, and methane management measures; obtain Site Audit Statements from NSW EPA-accredited Site Auditors where remediation or management measures are implemented
- Implement contamination pathway controls during earthworks and future development to prevent migration of identified COPCs to Cockle Creek and Lake Macquarie
- Ensure all contamination assessments comply with NSW EPA Guidelines and are prepared by a suitably qualified contaminated land consultant including any review and update of duty to report obligations under Section 60 of the *Contaminated Land Management Act 1997* at each DA-stage investigation and

preparation of Long-Term Environmental Management Plans (LTEMPs) where residual contamination risks remain. As per Lake Macquarie City Council's (LMCCs) requirements, an EPA certified consultant must be engaged where reports are submitted to the NSW EPA or where required by the consent authority

- All future works should be conducted by an appropriately qualified environmental consultant under the direction of a CEnvP-SC or equivalent accreditation. An accredited NSW EPA Auditor may be required to endorse the investigation, remediation/management and validation works where more sensitive land uses are proposed or where there are elevated risks to ecological receptors.

Pathway for Residential Development (Teralba Southgate)

Residential development at Teralba Southgate will require a standard but targeted contaminated land assessment and approval process, including:

- Detailed Site Investigation (DSI) involving intrusive assessment to fully characterise the nature and extent of contamination across the site in potentially affected media (i.e. ground gas and to a lesser extent soil, groundwater, surface water, sediment) with data comparison made to applicable land use criteria as per the NEPM (e.g. medium to high density residential).
- Remedial Action Plan (RAP) to define extents and media requiring remediation and/or management to render the site suitable for the proposed land uses (e.g. residential).
- Validation sampling and reporting to confirm remediation objectives have been achieved.
- Independent Site Audit by an accredited NSW EPA Auditor which is recommended for residential land use on a former mine site, to provide independent assurance of site suitability prior to development consent being granted.

This process will demonstrate that the land is suitable, or can be made suitable, for residential use in accordance with the NEPM (2013), EP&A Act, *State Environmental Planning Policy (SEPP, Resilience and Hazards) 2021* and relevant NSW EPA legislation and guidelines.

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1 Introduction

1.1 Introduction

Aurecon Australasia (Aurecon) was engaged by the Department of Planning, Housing and Infrastructure (DPHI) for a variety of engineering services to support the State-led rezoning of the Macquarie Coal Complex, herein referred to as “the Precinct” and “Teralba Southgate”. The Precinct and Teralba Southgate are located approximately 20 kilometres (km) southwest of Newcastle, in the Lake Macquarie region of New South Wales (NSW). It is noted that Teralba Southgate has been removed from the scope of the Master Plan and will not be pursued under a State Environmental Planning Policy (SEPP) rezoning process. Instead, Teralba Southgate is proposed to be rezoned under a Local Environmental Plan (LEP). A figure showing the location and extent of the Precinct and Teralba Southgate is provided in **Figure 1, Appendix A**.

The Precinct encompasses the former West Wallsend Colliery (WWC), the Macquarie Coal Preparation Plant (MCP), the former open-cut Westside Mine, and the former Teralba Colliery (Northgate site). Mining operations ceased progressively across the Precinct, with Teralba Colliery closing in 2001, Westside Mine ceasing operations in 2012, and West Wallsend Colliery and Macquarie Coal Preparation Plant concluding operations in 2016. Current activities are focused on long-term environmental management and rehabilitation in accordance with the Rehabilitation Management Plan (RMP 2022).

This Contamination Technical Assessment (CTA) builds on the Post Mining Land Use Engineering Analysis Report (Aurecon 2026) and has been developed to support refinement of the Master Plan and to inform DPHI's decision making as part of the rezoning process. It is noted that environmental and contamination investigations completed to date have been undertaken primarily to support mine closure obligations, rather than to establish suitability for alternative land uses. Accordingly, this assessment reinterprets existing contamination data within the context of the proposed rezoning and future land use scenarios shown in **Figure 2, Appendix A**.

1.2 Objectives

The objective of this CTA is to support refinement of the Master Plan and to inform DPHI's decision making as part of the rezoning process. Specifically, the objectives of this CTA are to:

- Provide a precinct-scale contaminated land evidence base to support rezoning and Master Plan land use compatibility, building on existing contamination assessments referenced through Enquiry by Design (EbD) engineering discussions
- Confirm whether contamination represents a material constraint to developable land verification for employment-led uses and identify where more sensitive land uses would require additional evidence and management
- Translate contamination findings into Master Plan-ready objectives, performance criteria and a clear staged investigation/remediation pathway to be implemented post rezoning.

1.3 Methodology

The scope of this CTA comprises five integrated components to achieve the rezoning objectives including:

- Desktop compilation including a review and synthesis of previous investigations and assessment documents
- Systematic constraints analysis including the identification of Areas of Environmental Concern (AECs) and assessment of remediation implications under different land use scenarios
- Development of a Conceptual Site Model (CSM) to integrate source pathway receptor (SPR) linkages across the Precinct and Teralba Southgate
- Identification of data gaps to inform recommendations for future investigation
- Development of a management framework to integrate contamination management into the future planning and development approval pathway, including staged investigation triggers, regulatory notification protocols, and interfaces with other engineering and environmental disciplines.

2 Planning context and limitations

2.1 Level of detail and exclusions

This CTA report has been prepared appropriate for a rezoning and Master Plan stage and is structured to support precinct-scale land use compatibility decisions and developable land verification for the Precinct and Teralba Southgate. This report builds on the existing information mainly from prior investigations undertaken across the Precinct and Teralba Southgate at various stages, primarily in support of mine closure obligations and reinterprets that data within the context of the proposed rezoning and future land use scenarios.

This CTA is not intended or scoped to deliver a development application (DA) level investigation, detailed remediation design, or construction validation for the Precinct and Teralba Southgate. The assessment has been prepared to inform the Structure Plan and development of a Master Plan and to support the DPHI's decision-making as part of the State-led rezoning process.

Typical items as part of the staged contamination risk assessment and assurance framework are excluded from this assessment and would need to be addressed progressively through the DA and approvals process as individual proposals are brought forward. These include the following:

- Preliminary Site Investigations (PSIs)
- Detailed site investigations (DSIs)
- Sampling Analysis and Quality Plans (SAQPs)
- Remedial Action Plan (RAP) or management plans relating to contamination including remediation costing.

2.2 Reliance on data and information

This report has been prepared based on documentation and information provided by DPHI and Glencore listed below and referenced as part of this report. Aurecon has not independently verified the accuracy or completeness of this information. of existing studies and data, except as otherwise noted in this report. The findings, advice, and recommendations presented in this report are reliant on this material being accurate and complete. Where data gaps or uncertainties have been identified, these are noted throughout this report and are discussed further in **Sections 5 to 7**.

The documents used in the preparation of this report are as follows:

- Post Mining Land Use – Engineering Analysis Report (DRAFT), prepared by Aurecon for DPHI
- Post Mining Land Use – Deliverability of Pilot Projects – Due Diligence Report – Macquarie Coal Complex, prepared by Aurecon in February 2026
- Preliminary statutory planning due diligence prepared by Patch Planning
- Insights obtained from internal workshops with DPHI, package leads and agencies
- Initial land use scenarios and sub precincts provided by GHD.

Contamination Discipline Reference Documents include:

- SLR (2016), Report on Intrusive Investigation of Fill Potentially Containing Furnace Slag – Coal Product Stockpile Areas, Macquarie Coal Preparation Plant, (Report No: OCCP-SLR-DOM00-CL-RPT-0002, dated 21 December 2016)
- SLR (2017), Contaminated Site Assessment Report – OCAL Complex (site-wide), (Report No: OCCP-SLR-DOM00-CL-RPT-0001, dated 22 February 2017)
- Coffey (2019), Contamination Site Assessment – Catchment Dams 1, 2 and 3, Macquarie Coal Preparation Plant, The Broadway, Killingworth NSW, (Report No: 754-NTLEN218485-R02, dated 16 April 2019)

- Coffey (2019), Contamination Site Assessment – Catchment Dams 1, 2 and 3, Macquarie Coal Preparation Plant, The Broadway, Killingworth NSW, (Report No: 754-NTLEN218485-R02-Rev1, dated 7 August 2019)
- Coffey (2019), Maturation Pond Remediation Validation Report – West Wallsend Colliery, The Broadway, Killingworth NSW, (Report No: 754-NTLEN218485-R03-Draft, dated 1 July 2019)
- Coffey (2019), Sediment and Surface Water sampling for presence of PFAS within Dam G, Westside Mine, Killingworth NSW, (Letter Report No: 754-NTLEN221349-L01a, dated 11 November 2019)
- Coffey (2020), Contaminated Site Assessment – Pit Top Catchment Dams, West Wallsend Colliery, The Broadway, Killingworth NSW, (Report No: 754-NTLEN218485-1-R03, dated 20 April 2020)
- Tetra Tech Coffey (TTC, 2021), Contamination Site Assessment – Teralba Southgate, 91 York Street, Teralba NSW (Report No: 754-NTLEN288358-R02, dated 16 November 2021)
- Coffey (2018), Dam Sediment Assessment – Teralba Southgate, 91 York Street, Teralba NSW (Report No: 754-NTLEN222379-R01, dated 13 November 2018)
- SLR (2016), Geochemistry and Spontaneous Combustion Investigation (Report No: OCCP-SLR-DOM00-GC-RPT-0001-Rev A, dated 8 October 2016)
- Coffey (2021), Human Health and Ecological Risk Assessment, Macquarie Coal Preparation Plant and West Wallsend Colliery (Report No: 754-NTLEN218485-1-R04, dated 24 February 2021)
- Tetra Tech Coffey (TTC, 2022), PFAS Human Health and Environmental Risk Assessment Teralba Southgate, 91 York Street, Teralba NSW (Report No: 754-NTLEN288358-R04a, dated 30 August 2022)
- SLR (2014), Preliminary Phase 2 Assessment – West Wallsend Colliery Pit Top (Report No: 610.14365-R1, dated 19 November 2014)
- GHD (2020), Teralba Power Station – ER, Summary Validation Report (Report No: 12518779, dated May 2020)
- RCA (2005), Phase 2 Environmental Site Assessment, Macquarie Coal Preparation Plant, Teralba (Report Ref: 4952-002/0, dated September 2005)
- Umwelt (2005), Phase 1 Contamination Assessment, Oceanic Coal Australia Limited (OCAL) (Report Ref: 1224/R47/V2, dated May 2005)
- WSP (2025), West Wallsend Tailing Dam Landform Finalisations Report (Report ID: PS216841-WSP-NTL-MNG-REP-0001, dated 3 October 2025)
- Macquarie Coal Glencore (2022), Macquarie Coal Complex Rehabilitation Management Plan Add in Macquarie Coal RMP (Dated August 2022).

3 Baseline analysis

This section presents the baseline environmental and contamination information for the Precinct and Teralba Southgate. The baseline analysis draws on a series of previous contamination investigations undertaken primarily to support mine closure obligations drawing on the existing evidence base compiled from previous investigations and publicly available information. Existing data and available information for the Precinct and Teralba Southgate have been reinterpreted within the context of the proposed rezoning and future land use scenarios identified in the Structure Plan (**Figure 1, Appendix A**). This section includes summary of the environmental setting information for the Precinct and Teralba Southgate, identifies key AECs and summary of previous remediation activities that form the basis for the CSM, and constraint assessment presented in **Section 4**.

3.1 Environmental setting

As outlined in **Section 1.1**, the Precinct is located approximately 20 kilometres southwest of Newcastle in Lake Macquarie, NSW. It comprises four sub-sites (West Wallsend Colliery, Macquarie Coal Preparation Plant, Westside Mine, and Teralba Colliery (Northgate)) and is currently zoned SP1, Special Activities (Mining), C2 Environmental Conservation, C3 Environmental Management and RU2 Rural Landscape pursuant to the *Lake Macquarie Local Environmental Plan 2014*. Teralba Southgate is currently zoned as Special Activities (Mining) and occupies an area of approximately 7 hectares (ha). Areas surrounding the Precinct and Teralba Southgate generally comprise residential communities and mixed-use industrial land uses. A summary of environmental features within the Precinct and Teralba Southgate is provided in **Table 3-1**.

Table 3-1 Summary of Environmental Setting

Environmental Feature	Description
Soil	A combined review of soil landscape information provided by eSPADE tool hosted by the NSW Department of Planning, Industry and Environment (DPE 2025) and previous investigations completed at the Precinct and Teralba Southgate indicates that surficial soils generally comprise sandy loam topsoil transitioning to sandy clay, with variable fill materials (1.9–8.5 m depth) containing coal fragments, concrete, brick and metal.
Geology	A review of geological information sourced from NSW Seamless Geology MinView (Version 2.5, released May 2025) paired with previous investigations indicates that the Precinct and Teralba Southgate is underlain by a mix of sandstone, conglomerate, shale, tuff, and coal seams.
Hydrogeology	Review of previous investigations indicates that a shallow unconfined groundwater underlies the Precinct and Teralba Southgate at approximate depths between 2 m below top of casing (btoc) and 15 m btoc. Groundwater quality was reported to be fresh to slightly brackish quality (EC 691–1,780 $\mu\text{S}/\text{cm}$) and generally flowed easterly toward Cockle Creek and Lake Macquarie.
Hydrology	Rainfall is expected to primarily be captured within on-site storage dams or flow to open drains in the north-east or south-east of the Precinct and Teralba Southgate. These drains eventually discharge to Cockle Creek located approximately 210 metres east of the Precinct. During periods of high or prolonged rainfall, it is expected that rainfall would flow overland and naturally diffuse to surrounding land.

Environmental Feature	Description
Acid Sulfate Soils (ASS)	Review of ASS risk mapping tool hosted by eSPADE (2025) indicates that the majority of the Precinct and Teralba Southgate is unlikely to comprise an ASS risk. However, areas characterised as having low ASS probability (1-3 m AHD, and >3 m AHD) occupy the sub precincts 2, 3, 6B and 6C. Areas comprising high probability (between 1–3 m AHD) are present along the far northern and eastern precinct margins within sub precincts 6b and 6D. Disturbed terrain classifications (elevation 1–4 m AHD) are mapped across portions of the eastern boundary, reflecting prior land disturbance that may have altered the vertical distribution and oxidation state of sulfidic materials. A figure showing the ASS probability for the Precinct and Teralba Southgate is provided in Figure 3, Appendix A. Site-specific ASS investigations completed in sub precinct 1 identified AASS and PASS in dam sediments, with Dams 2 and 3 confirming active acid generation (Coffey, 2019), though a subsequent assessment (Coffey, 2020) returned sulfide concentrations predominantly below action criteria. Maturation Pond sediments within sub precinct 1 were also confirmed acid-generating prior to dewatering, with rehabilitation involving lime neutralisation and relocation to a lined containment cell. A summary of site-specific ASS investigations undertaken at the Precinct and Teralba Southgate is provided in Appendix B.
Salinity	Based on salinity risk maps provided by NSW DCCEEW, the Precinct and Teralba Southgate is predominantly underlain by very high salinity hazard, covering the majority of sub precincts 1, 2, 3, 6A, 6B, 6C, 6D and 6E. Moderate salinity hazard is mapped across portions of the northern and eastern site extents, including parts of sub precincts 2, 6B, 6C, and 6D. Low salinity hazard is present across localised areas within sub precincts 1, 2, 3, 6A, 6B, 6C, and 6D, and along drainage corridors associated with Cockle Creek and its tributaries. A very high classification indicates significant potential for salt-affected soils and saline groundwater across the site, with implications for infrastructure design, vegetation establishment, and materials selection. A figure showing the salinity risks for the Precinct and Teralba Southgate is provided in Figure 4, Appendix A. Site-specific salinity data across the Precinct and Teralba Southgate is limited, with no assessment data available for the majority of domains. However, elevated salinity has been confirmed in sediments and groundwater in sub precinct 1, and surface water and shallow groundwater conditions in sub precinct 4 are consistent with saline or mineralised discharge. A summary of site-specific salinity data assessment available for the Precinct and Teralba Southgate is provided in Appendix B.

3.2 Areas of Environmental Concern

A total of 16 closure Domains (Areas of Environmental Concern, AECs) have been identified across the Precinct and Teralba Southgate as part of the Detailed Mine Closure Plan (DMCP). These Domains span the Westside Mine (Domains 1–5), West Wallsend Colliery (Domain 6), Macquarie Coal Preparation Plant (Domains 7–10), Teralba Northgate (Domain 11), site-wide infrastructure (Domain 12), Teralba Southgate (Domain 13), and other mine-owned and rural residential lands (Domains 14–16).

A figure showing the 16 closure domains is provided in Figure 5, **Appendix A**. A summary of the Domains and their former uses is provided in **Table 3-2**. Domain 15 falls outside the assessment area and detailed investigation is not warranted to achieve project objectives, however this domain has been included in **Table 3-2** for completeness.

Table 3-2 Summary of AECs (Closure Domains) across the Precinct and Teralba Southgate.

AEC	Name	Description / Former Uses
Domain 1	Westside Administration and Workshop Area	Former administrative and workshop facilities associated with the Westside Mine including former underground storage tanks (UST) and oil lubricant areas as well as building footprints and car park areas.

AEC	Name	Description / Former Uses
Domain 2	North Rehabilitation Area	Former rehabilitation area associated with the Westside Mine.
Domain 3	South-west Pit / Borrow Area	Former open-cut pit area and borrow pit associated with the Westside Mine.
Domain 4	Proposed Final Void / WWC Westside Void	Includes the Westside Final Void, Bioremediation Facility and Sediment Disposal Area.
Domain 5	Voluntary Conservation Areas	Areas of ecological conservation value partially associated with the Westside Mine.
Domain 6	WWC Pit Top Facilities including Dams	Former operational pit top facilities at West Wallsend Colliery, including workshop, coal handling plant, above-ground storage tanks, sedimentation dams and sewage treatment plant.
Domain 7	Tailings Dam	MCCP tailings storage facility.
Domain 8	Rejects Emplacement Area	MCCP rejects emplacement area.
Domain 9	MCCP Surface Facilities	Former coal processing plant, fuel storage facilities, product stockpile areas (coking and thermal coal), raw coal stockpile, rail loop and associated infrastructure.
Domain 10	Maturation Ponds	Former treated wastewater maturation ponds associated with the MCCP.
Domain 11	Teralba Northgate	Former Teralba Colliery Northgate surface facilities, including mine shafts (No. 3 and No. 4), workshop and laydown areas.
Domain 12	Infrastructure (Haul Road)	Main haul road connecting the Westside Mine to the MCCP.
Domain 13	Teralba Southgate	Former Teralba Colliery Southgate surface facilities, including workshop, laydown areas, methane plant, stormwater dam and above-ground infrastructure.
Domain 14	Other Mine Owned Lands	Miscellaneous mine-owned land parcels partially associated with the Westside Mine.
Domain 15 (Not assessed as part of this CTA)	Project Approval Area	Area subject to the original project approval. Falls outside the primary physical footprint of the MCC Study Area targeted for structure plan and zoning analysis. Not subject to the CSA.
Domain 16	Rural Residential Property	Rural residential properties associated with the Precinct. Not subject to the CSA due to low environmental risk.

3.3 Summary of Previous Investigations

Contamination investigations across the Precinct and Teralba Southgate have been undertaken progressively since 2005 spanning many of the 16 closure Domains. The investigations have been carried out by multiple consultants with varying objectives targeting different parts of the Precinct and Teralba Southgate. Environmental reports considered as part of this assessment are also listed in **Section 1.5**. A summary of investigations undertaken at each Domain is provided below and discussed further in **Appendix B**. Environmental reports considered as part of this assessment are also listed in **Section 1.5**.

RCA Australia (2005) undertook the earliest intrusive investigations across Domains 1 to 4. SLR (2017) completed a Precinct-wide assessment across Domains 1, 3, 4, and 6 and did not report human health exceedances under a commercial/industrial scenario, however detected ecological exceedances for heavy metals and TRH in fill soils. Coffey (2019 to 2020) targeted Domain 6 Catchment Dams, and reported additional ecological exceedances for zinc, copper, TRH, PAHs, and PFAS, as well as identified asbestos containing material (ACM) in shallow topsoil.

Domains 9 and 10 were assessed by SLR (2017) and Coffey (2018–2021). Partial validation of Domain 10 Maturation Pond remediation identified two residual exceedance locations and heavy metal exceedances and ASS were confirmed in Catchment Dam sediments. SLR (2016) identified potential impacts associated with slag in the coal stockpile areas of Domain 9. WSP (2025) assessed the Domain 7 TSF for geotechnical integrity and closure design only; no intrusive contamination data was included.

Domains 11 and 13 assessments confirmed ecological exceedances for heavy metals, TRH, PAHs, and PFAS in soils and groundwater, methane gas at Domain 11 mine shaft locations, and ACM at both domains. Domain 12 assessment by SLR (2017) identified ecological exceedances for zinc and TRH, and elevated selenium and uranium concentrations along the Haul Road.

3.4 Previous Remediation Activities

Progressive remediation activities have been undertaken at multiple domains within the Precinct and Teralba Southgate. A summary of the remediation activities undertaken at the Precinct and Teralba Southgate is outlined in **Table 3-3**.

Table 3-3 Summary of previous remediation activities

Domain ID	Domain Name	Remediation Activity
1	Westside Administration and Workshop Area	- Removal and disposal of oil/water separator - Identification and management of former UST area
2	North Rehabilitation Area	Aurecon understands that this domain was subject to remediation, however records have not been made available for review
4	Proposed Final Void / WWC Westside Void	Northwest corner used as bioremediation facility
6	WWC Pit Top Facilities including Dams	Sediments from the dam clean up were transported to Domain 4 for disposal
7	Tailings Storage Facility	- Removal of contour banks from the northern outer TSF wall is currently underway - Fine ash tailings placement onto the TSF crest is ongoing to enable access for capping works
10	Maturation Ponds	- Removal of asbestos pipe work - Removal of ASTs associated with anti-flocculent infrastructure - Removal of ACM baffle wall from MP05 - MP01-MP05 - Excavation of sediment and approximately 0.2m of underlying soil material

3.5 Data gaps

While several investigations have been undertaken at precinct, Aurecon have identified a variety of data gaps and uncertainties associated with the level of assessment completed to date. The most common data gaps include absence of surface or groundwater data across multiple domains, incomplete or absent ASS characterisation across the majority of the Precinct and Teralba Southgate, unassessed building footprints and inaccessible structures in Domains 1 and 11, incomplete surface water and sediment quality assessment in several domains, and extents of buried asbestos pipework in Domains 6 and 10 has not been fully defined. A general data gap across multiple domains is the use of superseded hydrocarbon assessment criteria, with hydrocarbon data previously assessed against NEPM 1999 rather than the current NEPM 2013. These data gaps are summarised in **Appendix C** and will need to be progressively resolved through the staged DA-stage investigation programs.

4 Conceptual Site Model

4.1 Sources, pathways, and receptors

Risks associated with potential site contamination are based on evaluating the linkages between sources, pathways and receptors (SPRs based) on site specific environmental setting, potential design strategies, and proposed land use scenarios. The development of a conceptual site model (CSM) for the Precinct and Teralba Southgate facilitates an evaluation of the source-pathway-receptor relationships to assist in identifying potential contamination risks based on the nature of the contamination sources, degree of exposure of each receptor to a source, and the sensitivity of the likely receptors. A general description of the CSM components required to evaluate contamination risk is provided in **Table 4-1**.

Table 4-1 Summary of key CSM components

Term	Description
Source	A substance that has the potential to cause harm to human or environmental receptors. Sources can be primary (at the original point of release) or secondary (at a point further down-gradient from the primary source following temporary storage and / or transformation) and can be either direct (i.e. emanating from a localised area) or diffuse (i.e. widespread) in nature. In a broader sense, sources can also include particular ground conditions that may impact on construction materials such as aggressive soils (i.e. ASS).
Pathway	The route by which the source is brought into contact with the receptor. This can include the transport of contamination via water (surface and groundwater), airborne dust, vapours, and particulate-matter deposition. It can also include solubility or chemical transformation from a relatively harmless chemical species to one that is more toxic.
Receptor	Human beings, other living organisms, and built structures that could be affected by the contamination source. A receptor will only be affected if an exposure pathway from the source to the receptor is complete. Groundwater and surface water systems can be considered as receptors as their quality is regulated by statutory bodies, as well as being potential pathways for contaminant migration to other receptors. In a broader sense, a receptor can be building or construction materials from the presence of aggressive soil conditions.

4.2 Domain Constraints Analysis

This section presents a preliminary constraints analysis for each domain presented in **Section 3.2**. The constraints analysis provides a structured, risk-based assessment for each domain based on available data and the current structure plan. The analysis is intended to support the rezoning and land use suitability assessments at the Precinct and Teralba Southgate and identify where further investigation or remediation is required to support or progress the rezoning. The constraint rating for each domain was derived by combining a consequence qualification (**Section 4.2.1**) with a likelihood qualification (**Section 4.2.2**) using the constraint determination matrix (**Section 4.2.3**).

4.2.1 Consequence qualification

Consequence is defined as the magnitude of potential adverse planning or environmental outcome that could result if a constraint were it to materialise in the context of the proposed rezoning strategy. Consequences were qualified across three categories that reflect the key considerations relevant to a planning proposal and land use suitability assessment under the EP&A Act and CLM Act frameworks. A summary of potential consequences and their associated ratings is presented in **Table 4-2**.

Table 4-2 Consequence qualifications and ratings

Consequence rating	Human health and Land Use Suitability	Ecological Environmental Values	Remediation Feasibility
Severe	Site conditions incompatible with proposed sensitive land use	Irreversible impact on threatened ecological community	Remediation feasibility uncertain under rezoning or land use scenario.
High	Health risks present rezoning suitability likely determined by defined detailed assessment or remediation.	Constraint may materially affect ecological values under rezoning or land use scenario.	Remediation feasible but requires defined scope, programme, and validation framework.
Moderate	Exceedances or possible data gaps may preclude proposed zoning or land use subject to targeted investigation or remediation.	Potential ecological impact manageable through targeted assessment, controls, or approval conditions.	Remediation feasible and likely achievable within standard programme.
Mild	Minor human health exceedances or data gaps are unlikely to be material.	Low potential for impact to ecological values; addressable via standard planning conditions.	Manageable through standard site preparation or minor targeted remediation at DA or construction stage.
Negligible	Contaminant concentrations are likely or confirmed to be below screening criteria.	No measurable or expected impact on ecological values.	No remediation expected; contaminant concentrations present no impediment to the rezoning.

4.2.2 Likelihood qualification

Likelihood is defined as the probability that a potential constraint will occur, or be confirmed as present, under the proposed land use scenario. Likelihood was qualified based on the weight of available evidence, including historical land use data, existing analytical results, site observations, and any identified data gaps. A summary of the adopted likelihood ratings is presented below:

- **Rare** – No credible evidence supports the presence of contamination constraints under the proposed land use. Occurrence of elevated contaminants may be present under exceptional circumstances, however is not supported by available data
- **Unlikely** – Available data does not support the presence of contamination constraints. Presence of elevated contaminants under proposed land use scenarios is plausible but not expected
- **Possible** – Contamination constraints may be present or may occur based on identified potential sources or pathways
- **Likely** – Strong evidence or credible site history suggests contamination constraints are present or will manifest during redevelopment
- **Almost Certain**– Contamination constraints are expected to be present based on confirmed site history, analytical data, or physical observation. Occurrence is effectively certain under the proposed land use and/or development scenario.

4.3 Constraint determination

A constraint rating for each domain was determined by combining the consequence and likelihood ratings using the matrix in **Table 4-3**. The matrix applies a conservative approach consistent with the range of proposed land uses, with greater weight given to consequence where data gaps exist and receptor sensitivity is high (e.g. residential and conservation precincts). Where a domain transects multiple land use zones, the more sensitive land use was adopted to derive the constraints rating.

Table 4-3 Domain constraint determination matrix

	Likelihood				
Consequence	Rare	Unlikely	Possible	Likely	Almost Certain
Severe	Low	Moderate	High	Very High	Very High
High	Low	Low to Moderate	Moderate to High	High to Very High	Very High
Moderate	Negligible to Low	Low	Moderate	Moderate to High	High
Mild	Negligible	Low	Low	Low to Moderate	Moderate
Negligible	Negligible	Negligible	Negligible to Low	Low	Low

Aurecon has defined the constraints ratings outlined in Error! Reference source not found. as the following:

- **Negligible** – No known or likely constraint to the proposed rezoning or land use scenario. Contaminant concentrations and site conditions are likely consistent with background concentrations or below thresholds and do not represent a planning, regulatory, or environmental risk
- **Low** – Minor or manageable constraint that can be addressed through standard construction and environmental management practices. No material impact on rezoning or redevelopment, subject to further assessment and standard regulatory compliance
- **Moderate** – Moderate constraint that warrants further assessment and consideration in future planning, design or construction phases. Risk-based management measures, development controls, or targeted remediation are likely required
- **High** – Significant constraint requiring targeted investigation and/or remediation prior to or concurrent with rezoning or redevelopment. The proposed land use may be achievable subject to a defined remediation or risk management strategy and, where applicable, a Site Audit Statement
- **Very High** – Fundamental constraint to proposed rezoning or redevelopment program. Contamination or environmental conditions are likely incompatible with the proposed land use without significant remediation, risk management, or design modification.

4.4 Preliminary conceptual site model

As outlined in **Section 3.2**, a number of domains were identified across the Precinct and Teralba Southgate which contain a variety of contamination sources. As such, these may represent potential constraints in context to the land use scenarios outlined in the current structure plan (**Figure 2, Appendix A**). A preliminary CSM is presented in **Table 4-4** and includes a summary of the domains located at the Precinct and Teralba Southgate, and potential SPR linkages and constraints ratings. The COPCs listed reflect findings from previous investigations and are assessed in the context of the proposed future land uses. Where no intrusive investigation data was available for the Precinct, COPCs were determined based available site history information. It should be noted that Domain 15 is located external to the Precinct and Teralba Southgate and has therefore been excluded from **Table 4-4**. Constraint rating qualifications and associated confidence ratings are included in **Appendix D**.

Table 4-4 Preliminary CSM for the Precinct and Teralba Southgate

Domain ID	Domain Name	Structure Plan zone IDs	Proposed Zoning	Inferred NEPM Land Use Category	Key Contamination Sources	Primary CoPCs	Potentially impacted media	Potential pathways	Potential receptors	Risk Rating
1	Westside Administration and Workshop Area	2, 6b	Killingworth (Employment Core), Conservation + Open Space	HIL-D (Commercial/Industrial) EILs (Conservation)	- Former USTs and oil/lubricant storage areas - Vehicle fleet maintenance area	TRH/BTEXN, PAH, Heavy Metals, Asbestos, Phenols, PCBs, PFAS	- Soil - Groundwater - Surface water (Dam G) - Sediments (Dam G) - Soil vapour	- Direct dermal contact with contaminated media - Inhalation of contaminated media - Infiltration to groundwater - Surface water runoff - Vapour intrusion/accumulation in enclosed spaces	- Current and future users of the Precinct - Construction and maintenance workers - Ecological receptors including flora and fauna at the Precinct and surrounding areas (Cockle Creek, Lake Macquarie, and wetlands)	Moderate
2	North Rehabilitation Area	2, 6b	Killingworth (Employment Core), Conservation + Open Space	HIL-D (Commercial/Industrial) EILs (Conservation)	Historical association with Westside Mine operations, potential for residual hydrocarbon and metal impacts from adjacent activities.	TRH/BTEXN, PAH, OCP/OPP, Heavy Metals, Asbestos, PFAS	- Soil - Groundwater - Surface water - Soil vapour	- Direct dermal contact with contaminated media - Inhalation of contaminated media - Infiltration to groundwater - Surface water runoff - Vapour intrusion/accumulation in enclosed spaces	- Current and future users of the Precinct - Construction and maintenance workers - Ecological receptors including flora and fauna at the Precinct and surrounding areas (Cockle Creek, Lake Macquarie, and wetlands)	Low
3	Borrow Pit Area	3, 6a	West Lake (Mixed Use), Conservation + Open Space	HIL-B (Residential) HIL-D (Commercial/Industrial) EILs (Conservation)	- Borrow pit - Broader areas outside the borrow pit - Potential for residual hydrocarbon impacts from bioremediation pad operations.	TRH/BTEXN, PAH, Heavy Metals, Phenols, Asbestos, pH	- Soil - Surface water - Groundwater - Soil vapour	- Direct dermal contact with contaminated media - Surface water runoff - Infiltration to groundwater - Inhalation of contaminated media - Vapour intrusion/accumulation in enclosed spaces	- Construction and maintenance workers - Current and future users of the Precinct - Ecological receptors including flora and fauna at the Precinct and surrounding areas (Cockle Creek, Lake Macquarie, and wetlands)	Low
4	Westside Final Void / WWC Westside Void	3	West Lake (Mixed Use), Conservation + Open Space	HIL-B (Residential) HIL-D (Commercial/Industrial) EILs (Conservation)	- Bioremediation facility operations - Sediment disposal from Domain 6 dam remediation	TRH/BTEXN, PAH, Heavy Metals, Phenols, Nutrients, pH	- Soil - Sediments - Surface water - Groundwater	- Surface water runoff - Direct dermal contact with contaminated media - Infiltration to groundwater	- Current and future users of the Precinct - Construction and maintenance workers - Ecological receptors including flora and fauna at the Precinct and surrounding areas (Cockle Creek, Lake Macquarie, and wetlands)	Moderate

Domain ID	Domain Name	Structure Plan zone IDs	Proposed Zoning	Inferred NEPM Land Use Category	Key Contamination Sources	Primary CoPCs	Potentially impacted media	Potential pathways	Potential receptors	Risk Rating
5	Voluntary Conservation Areas	6b	Conservation + Open Space	EILs (Conservation)	- ASS risk mapped across low-lying portions of the domain. - Potential for residual impacts from adjacent Westside Mine operations.	TRH/BTEXN, PAH, OCP/OPP, Heavy Metals, Asbestos, PFAS	- Soil - Groundwater - Surface water	- Direct dermal contact with contaminated media - Inhalation of contaminated media - Infiltration to groundwater - Surface water runoff	- Current and future users of the Precinct - Construction and maintenance workers - Ecological receptors including flora and fauna at the Precinct and surrounding areas (Cockle Creek, Lake Macquarie, and wetlands)	Low
6	WWC Pit Top Facilities	2	Killingworth (Employment Core), Conservation + Open Space	HIL-D (Commercial/Industrial) EILs (Conservation)	- Workshop and coal handling operations - Sedimentation dams receiving temporary coal stockpile runoff - Sewage treatment plant - Methane drainage well - Fibrous asbestos in surface soils and underground asbestos cement pipes	TRH/BTEXN, PAH, Heavy Metals, Phenols, Nutrients, Asbestos, OCP/OPP, PCBs, PFAS, pH, Ground gas (CO, CH ₄ , H ₂ S, CO ₂)	- Soil - Sediments - Surface water - Groundwater - Soil Vapour - Ground Gas - Ambient Air	- Direct dermal contact with contaminated media - Inhalation of contaminated media - Surface water runoff - Infiltration to groundwater - Vapour intrusion - Accumulation of HGGs in enclosed spaces	- Construction and maintenance workers - Current and future users of the Precinct - Ecological receptors including flora and fauna at the Precinct and surrounding areas (Cockle Creek, Lake Macquarie, and wetlands)	Moderate
7	Tailings Storage Facility	1, 6e	Cockle Creek (Employment Core), Conservation + Open Space	HIL-D (Commercial/Industrial) EILs (Conservation)	- Tailings storage and associated leachate - Potential acid mine drainage (AMD) - Dam seepage to groundwater and surface water - Geochemical characteristics of mine waste (PAF, spontaneous combustion, salinity)	TRH/BTEXN, PAH, Heavy Metals, PFAS, pH, Nutrients, Ground gas (CO, CH ₄ , H ₂ S, CO ₂)	- Soil - Sediments - Groundwater - Surface water - Soil Vapour - Ground gas - Ambient Air	- Direct dermal contact with contaminated media - Inhalation of contaminated media - Infiltration to groundwater - Surface water runoff - Vapour Intrusion - Accumulation of HGGs in enclosed spaces	- Current and future users of the Precinct - Construction and maintenance workers - Ecological receptors including flora and fauna at the Precinct and surrounding areas (Cockle Creek, Lake Macquarie, and wetlands)	Moderate to high ¹
8	MCCP Rejects Emplacement Area	1	Cockle Creek (Employment Core), Conservation + Open Space	HIL-D (Commercial/Industrial) EILs (Conservation)	- Coal reject material emplacement - Proximity to mine shafts and coal stockpiles (ground gas risk) - Slag materials	TRH/BTEXN, PAH, Heavy Metals, Phenols, Nutrients, pH, Ground gas (CO, CH ₄ , H ₂ S, CO ₂)	- Sediments - Surface water - Groundwater - Soil Vapour - Ground gas - Ambient Air	- Surface water runoff - Direct dermal contact with contaminated media - Inhalation of contaminated media - Infiltration to groundwater - Vapour Intrusion - Accumulation of HGGs in enclosed spaces	- Construction and maintenance workers - Ecological receptors including flora and fauna at the Precinct and surrounding areas (Cockle Creek, Lake Macquarie, and wetlands)	Low

Domain ID	Domain Name	Structure Plan zone IDs	Proposed Zoning	Inferred NEPM Land Use Category	Key Contamination Sources	Primary CoPCs	Potentially impacted media	Potential pathways	Potential receptors	Risk Rating
9	Washery and Workshop Area	1, 6d	Cockle Creek (Employment Core), Conservation + Open Space	HIL-D (Commercial/Industrial) EILs (Conservation)	- Fuel storage facilities and dispensing areas - Catchment Dams 1–3 and ponds receiving temporary coal stockpile runoff. - Coal stockpile areas (incl. ROM, CPP, Emergency SP areas) - Washery and Workshop - Rail Loop - Grave Yard - Haul Road - Underground asbestos cement pipes - PASS/ASS - PFAS in groundwater	TRH/BTEXN, PAH, Heavy Metals, Phenols, OCP/OPP, PCB, Asbestos, Nutrients, PFAS	- Surface soil - Near surface soil - Groundwater - Surface water - Sediment	- Direct dermal contact with contaminated media - Inhalation of contaminated media - Surface water runoff - Infiltration to groundwater	- Current and future users of the Precinct - Construction and maintenance workers - Ecological receptors including flora and fauna at the Precinct and surrounding areas (Cockle Creek, Lake Macquarie, and wetlands)	Moderate
10	Maturation Ponds	1, 6d	Cockle Creek (Employment Core), Conservation + Open Space	HIL-D (Commercial/Industrial) EILs (Conservation)	- Maturation Pond areas (residual sediment contamination post-remediation) - Former wastewater maturation - PASS/ASS - Underground asbestos cement pipes - PFAS in surface water	TRH/BTEXN, PAH, Heavy Metals, Phenols, OCP/OPP, Nutrients, pH, PFAS	- Groundwater - Surface Soil - Sediments	- Direct dermal contact with contaminated media - Inhalation of contaminated media - Infiltration to groundwater - Surface water runoff	- Current and future users of the Precinct - Construction and maintenance workers - Ecological receptors including flora and fauna at the Precinct and surrounding areas (Cockle Creek, Lake Macquarie, and wetlands)	Low
11	Teralba Northgate	1, 6d	Cockle Creek (Employment Core), Conservation + Open Space	HIL-D (Commercial/Industrial) EILs (Conservation)	- Mine shafts No. 3 and 4 (methane emissions confirmed exceeding LEL); - Workshop and laydown areas - Asbestos-containing materials (ACM) in fill from historical building demolition - Sewage contamination near septic tank area.	TRH/BTEXN, PAH, Heavy Metals, Phenols, OCP/OPP, PCBs, Nutrients, Ground gas (CO, CH ₄ , H ₂ S, CO ₂), PFAS, Asbestos, Biologicals	- Surface soil - Groundwater - Surface water - Soil Vapour - Ground gas - Ambient Air - Sediment	- Direct dermal contact with contaminated media - Inhalation of contaminated media - Surface water runoff - Infiltration to groundwater - Vapour intrusion - Accumulation in enclosed spaces	- Current and future users of the Precinct - Construction and maintenance workers - Ecological receptors including flora and fauna at the Precinct and surrounding areas (Cockle Creek, Lake Macquarie, and wetlands)	Moderate
12	Infrastructure (Haul Road)	1, 2, 5, 6a, 6b, 6c, 6f	Cockle Creek (Employment Core), Killingworth (Employment Core), Barnsley West (Employment Core), Conservation + Open Space	HIL-D (Commercial/Industrial) EILs (Conservation)	- Haul road operations (hydrocarbons, metals in fill); - PASS/ASS confirmed in underlying natural soils - Anomalous selenium and uranium enrichment at a Haul Road section	TRH/BTEXN, PAH, Heavy Metals, Phenols, PFAS, OCP/OPP, pH	- Soil - Surface water - Groundwater	- Direct dermal contact with contaminated media - Inhalation of contaminated media - Surface water runoff - Infiltration to groundwater	- Current and future users of the Precinct - Construction and maintenance workers - Ecological receptors including flora and fauna at the Precinct and surrounding areas (Cockle Creek, Lake Macquarie, and wetlands)	Low

Domain ID	Domain Name	Structure Plan zone IDs	Proposed Zoning	Inferred NEPM Land Use Category	Key Contamination Sources	Primary CoPCs	Potentially impacted media	Potential pathways	Potential receptors	Risk Rating
13	Teralba Southgate	Not Applicable	Teralba (Mixed Use)	HIL-B (Residential)	- Workshop and laydown areas where hydrocarbon impacts were confirmed incl. in hardstand area - methane plant area (including asbestos in fill near methane plant) - Stormwater dam - Underground mine workings (methane) - PFAS in groundwater	TRH/BTEXN, PAH, Heavy Metals, Asbestos, Phenols, OCC/OCP, Nutrients, Methane gas, PFAS, pH	- Soil - Groundwater - Surface water - Sediments - Soil Vapour - Ground gas - Ambient Air	- Direct dermal contact with contaminated media - Inhalation of contaminated media - Infiltration to groundwater - Surface water runoff - Vapour Intrusion - Accumulation in enclosed spaces	- Current and future users of Teralba Southgate - Construction and maintenance workers - Ecological receptors including flora and fauna at the Precinct and surrounding areas (Cockle Creek, Lake Macquarie, and wetlands)	Moderate to high ²
14	Other Mine Owned Lands	1, 2, 3, 5, 6a, 6b, 6c, 6d, 6e, 6f	Cockle Creek (Employment Core), Killingworth (Employment Core), West Lake (Mixed Use), Barnsley West (Employment Core), Conservation + Open Space	HIL-B (Residential) HIL-D (Commercial/Industrial) EILs (Conservation)	- Historical association with mine operations across multiple sub-areas. - ASS risk mapped in low-lying areas	TRH/BTEXN, PAH, Heavy Metals, Asbestos, PFAS, Nutrients, Phenols	- Soil - Groundwater - Surface water - Sediments	- Direct dermal contact with contaminated media - Inhalation of contaminated media - Infiltration to groundwater - Surface water runoff	- Current and future users of the Precinct - Construction and maintenance workers - Ecological receptors including flora and fauna at the Precinct and surrounding areas (Cockle Creek, Lake Macquarie, and wetlands)	Low
16	Rural Residential Property	3, 6a, 6b	West Lake (Mixed Use), Conservation + Open Space	HIL-B (Residential) EILs (Conservation)	No confirmed sources identified	TRH/BTEXN, PAH, Heavy Metals, Asbestos, PFAS	- Soil - Groundwater - Surface water - Sediments	- Direct dermal contact with contaminated media - Inhalation of contaminated media - Infiltration to groundwater - Surface water runoff	- Current and future users of the Precinct - Construction and maintenance workers - Ecological receptors including flora and fauna at the Precinct and surrounding areas (Cockle Creek, Lake Macquarie, and wetlands)	Negligible

Table notes:

¹ The 'moderate-to-high' rating reflects the absence of any contaminated land investigation of the TSF to date (soil, sediment, leachate, surface water and groundwater), rather than confirmed contamination at this domain. In the absence of site-specific data at this domain, a conservative rating has been adopted consistent with standard contaminated land assessment practice, pending targeted investigation.

² The 'moderate-to-high' rating primarily reflects the limited scope of ground gas monitoring completed to date, rather than a confirmed high level of hazard. This is on the basis that mine shafts remain open and capped at Teralba Southgate and previous ground gas investigations were not undertaken in accordance with NSW EPA Assessment and Management of Hazardous Ground Gas (NSW EPA 2020) guidelines.

5 Land Use Capability

The following sections present key findings as they relate to the current structure and master plan, interpreting available contamination data and information including analytical results, data gaps, and management requirements necessary to support the proposed employment-led uses. Where contamination is known or potentially present, recommendations are provided to guide remediation priorities and validation requirements prior to development. A figure showing the locations of the domains as they relate to the structure plan is provided in Figure 7, **Appendix A**.

Throughout the following sections, the term ‘investigation level’ is used which refer to Tier I screening values included in the NEPM 2013 (or its preceding version) under commercial industrial land use, or other guidance documents such as NEMPv3.1 and ANZG 2018. Tier I screening values are generic concentrations, exceedance of which indicate a potential risk could be present in a certain area. Exceedance of investigation levels or Tier I screening values do not automatically require remediation or management, but rather further investigation and additional tiers of risk assessments are required. It is possible that quantitative human health and/or ecological risk assessments can be developed to establish site specific levels that inform the necessary extent of remediation/management required to render a particular area as suitable for an intended land use.

5.1 Sub precinct 1 – Cockle Creek

Exceedances of investigation levels have been confirmed across multiple contaminants and domains within the Cockle Creek sub-precinct. Concentrations of zinc, copper, selenium, uranium, PFAS, nutrients and TRH in soil were reported above the ecological assessment criteria adopted during previous investigations. Exceedances of human health criteria for industrial/commercial land use were not detected in soil at this sub-precinct, however fragments of asbestos in fill were identified in fill materials (Domain 11). Detectable methane gas emissions were also recorded from mine shafts, indicating a potential ground gas hazard.

While this sub-precinct has been subject to a variety of previous investigations and localised remediation (Domain 10), data gaps and uncertainties exist within this sub-precinct including Domains 7 and 14, ASS characterisation, asbestos pipework, and targeted assessment of ground gas, soil/sediment, surface water, and groundwater. Domain 7 may represent the most significant constraint on the basis as an intrusive investigation has not been undertaken for this domain. These data gaps impact the ability to confidently verify land suitability for employment-led uses without additional evidence and management measures.

Based on available data, and with consideration to identified data gaps, known or potential contamination is unlikely to preclude commercial/industrial rezoning but requires targeted investigation of soil, sediment, groundwater, surface water and soil vapour/ground gas to confirm land suitability. Heavy industrial uses, as currently proposed, are considered compatible with the current understanding of the contamination profile at this sub-precinct provided appropriate assessment and management controls are implemented during future design and construction phases. Implementation of these management measures will enable the Precinct to support employment-led development while protecting environmental values in Cockle Creek and Lake Macquarie.

5.2 Sub precinct 2 – Killingworth

Exceedances of investigation levels have been confirmed across multiple domains within the Killingworth sub-precinct. Concentrations of TRH (C10–C16) in fill were detected above adopted ecological assessment criteria in Domain 1 which were associated with historical fuel storage and former UST areas. Exceedances of EILs for zinc and TRH were also confirmed in Domain 2, and anomalous enrichment of selenium and uranium was identified at a targeted location in Domain 12. Exceedances of human health investigation levels were not identified within this sub-precinct.

Data gaps remain within this sub-precinct including Domains 2 and 14 which have not been subject to intrusive investigations. Domain 3 is lacking data outside the borrow pit area, and limited assessment of ASS within this sub-precinct. The absence of intrusive data for Domains 2 and 14 represents the most significant constraint on land suitability determination.

Based on available data, known or potential contamination is unlikely to preclude commercial/industrial rezoning of the Killingworth sub-precinct but requires targeted investigation of soil, groundwater, surface water, soil vapour, and ASS across all domains to confirm land suitability for the proposed light industrial use. Previously reported hydrocarbon data also require reassessment against current NEPM 2013 criteria. Implementation of appropriate assessment and management measures during future investigation and design phases will enable the sub-precinct to support employment-led development.

5.3 Sub precinct 3 – West Lake

Exceedances of ecological assessment criteria have been confirmed at Domain 4 within this sub-precinct. Concentrations of zinc and nutrients in surface sediments were recorded above ecological screening levels on the north-west margin of the final void of Domain 4. Hydrocarbons and nutrients also exceeded ESLs in sediment samples from this domain. Exceedances of ecological or human health assessment criteria were not detected in the borrow pit area within Domain 3, indicating that soil conditions in this domain are suitable for the proposed commercial-industrial land use based on available data.

Data gaps constrain land suitability determination across this sub-precinct. Domain 3 lacks intrusive investigation data outside the borrow pit area and requires hydrocarbon data reassessment under NEPM 2013. Domain 4 presents the most significant constraint, with limited available information on bioremediation operations and surface water quality retained in the void, insufficiently investigated peripheral areas, and submerged sediments inaccessible for assessment. While Domains 14 and 16 are considered to represent low risk areas, they have not been subject to intrusive investigations and remain data gaps.

Based on available data and information, known or potential contamination is unlikely to preclude commercial-industrial rezoning of West Lake; however, targeted investigation of soil, sediment, groundwater, and surface water is required to confirm land suitability for the proposed tourism and resort uses. The ecological exceedances identified in Domain 4 require assessment and management to ensure compatibility with the proposed mixed-use development. Additional investigations of all domains within this sub-precinct are necessary to address data gaps and verify that these areas do not present unmanageable constraints. Surface water within the void (Domain 4) should particularly be subject to detailed assessment given it's proposed to be used for recreational purposes. Implementation of additional assessment programs and management controls for this sub-precinct would be necessary to support the proposed rezoning.

5.4 Teralba Southgate

Exceedances of investigation levels have been confirmed for multiple contaminants within Teralba Southgate. Domain 13 (Teralba Southgate) exhibits confirmed exceedances of ecological criteria for zinc, TRH, and benzo(a)pyrene (BaP) in soil, and groundwater exceedances for hexavalent chromium, nickel, zinc, and PFAS. Multiple exceedances of sediment criteria have also been recorded for TRH, cadmium, copper, lead, mercury, nickel, and zinc. Other identified impacts included the confirmed presence of chrysotile asbestos in fill material near the methane plant, and methane gas concentrations exceeding the lower explosive limit (LEL) detected during drilling. However, exceedances of human health criteria for commercial/industrial land use were not detected during previous investigations, indicating the potential suitability of this site for industrial use from a human health perspective.

Data gaps and uncertainties remain within this site that limit the ability to fully characterise the contamination profile and assess suitability for the proposed mixed-use development including high-density residential, recreation, and commercial-industrial uses. Further assessment of soil and groundwater is required, particularly sampling beneath the workshop, which has not been completed. Asbestos delineation sampling has not been undertaken to confirm the full extent of ACM-impacted materials. Surface water sampling has not been completed, and detailed ground gas assessments remain outstanding. Additionally, the proposed rezoning to high-density residential mixed-use introduces more sensitive land use scenarios than the current industrial classification, requiring assessment against residential and public open space criteria rather than commercial/industrial for certain exposure pathways.

Based on available data and with consideration to identified data gaps and limitations, known or potential contamination presents a constraint to the proposed high-density residential mixed-use rezoning of this sub-precinct. While the absence of human health criteria exceedances under commercial/industrial land use confirms likely suitability for continued light industrial purposes, the proposed rezoning to include high-density residential and recreational uses introduces more sensitive exposure scenarios that require targeted investigation and risk assessment. The site is likely suitable for the proposed commercial-industrial and recreation components of the mixed-use development, provided appropriate assessment and management controls are implemented during future design and construction phases. However, the residential component requires completion of targeted investigation of ground gas, and to a lesser extent soil, groundwater, and surface water to confirm land suitability under residential exposure scenario and to delineate the PFAS plume extent. Implementation of management measures including potential capping of PFAS-impacted areas, groundwater monitoring, asbestos delineation, and methane management will be necessary to enable the this site to support the proposed mixed-use development while protecting environmental values in Cockle Bay and Lake Macquarie and ensuring human health protection for future residential occupants.

As outlined in Section 1.1, through the rezoning process, land at Teralba Southgate is intended for residential purpose. For residential development to proceed, targeted contamination investigations will be required to demonstrate that Teralba Southgate is suitable, or can be made suitable, for residential land use consistent with the requirements of the NEPM 2013, EP&A Act, *State Environmental Planning Policy (SEPP, Resilience and Hazards) 2021* and relevant NSW EPA legislation and guidelines. This will typically require a staged investigation and reporting pathway involving the preparation of the following documents:

- DSI involving intrusive assessment to fully characterise the nature and extent of contamination across the site in potentially affected media (i.e. ground gas, soil, groundwater, surface water, sediment) with data comparison made to applicable land use criteria as per the NEPM (e.g. medium to high density residential).
- RAP to define extents and media requiring remediation and/or management to render the site suitable for the proposed land uses (e.g. residential).
- Validation sampling and reporting to confirm remediation objectives have been achieved.
- Site Audit Report by a NSW EPA accredited site auditor, which is recommended for residential land use on a former mine site, to provide independent assurance of site suitability prior to development consent being granted.

5.5 Sub precinct 5 – Barnsley West

Limited exceedances of investigation levels have been detected in domains associated with this sub-precinct. Domain 12 occupies a small portion of this sub-precinct and is known to contain exceedances of ecological criteria for zinc and TRH in fill materials. However, human health criteria exceedances have not been recorded for this domain under a commercial/industrial land use. Domain 14 has not been subject to intrusive investigations and as such, data and information is not available for interpretation.

Data gaps limit a detailed understanding of the contamination profile for this sub-precinct. ASS may underlie this sub-precinct, and intrusive data is required to determine whether it represents a constraint to future development. While this sub-precinct lacks available intrusive data, potential contamination is unlikely to preclude commercial and industrial land use given the lower receptor sensitivity associated with this land use category. Implementation of a targeted investigation program including soil and groundwater sampling during future development applications will enable confirmation of site suitability for commercial/industrial development while protecting worker health and environment values.

5.6 Sub precinct 6 –Environmentally Sensitive Areas

5.6.1 Sub precinct 6a

Limited exceedances of investigation levels have been confirmed in multiple domains associated with this sub-precinct. Domain 3 exhibits no exceedances of human health or ecological investigation levels. Domain 12 presents confirmed exceedances of ecological criteria for zinc and TRH in fill materials, with anomalous enrichment of selenium and uranium identified at a targeted haul road location. Domains 14 and 16 have not been subject to intrusive investigations. No human health criteria exceedances have been recorded within this sub-precinct based on available data.

Data gaps constrain full characterisation of the contamination profile across this sub-precinct. Domain 3 lacks investigation data outside the borrow pit area and requires hydrocarbon data reassessment under NEPM 2013. Domain 12 presents minor data gaps including uncharacterised selenium and uranium enrichment, and data has not been collected for Domains 14 and 16.

Known or potential contamination is unlikely to preclude conservation rezoning of Conservation Area 6a. The absence of human health exceedances and limited ecological findings confirm broad compatibility with conservation land use. Confirmation of suitability requires targeted soil investigation at this sub-precinct to close identified data gaps and further evaluate potential ecological risks to enable the Precinct to support conservation land use while protecting environmental values and Lake Macquarie.

5.6.2 Sub precinct 6b

Exceedances of investigation levels have been confirmed across multiple contaminants and domains associated with this sub-precinct. Domain 1 confirms EIL exceedances for TRH F2 (C10–C16) in fill soils associated with fuel and lubricant storage, vehicle refuelling, and former UST areas. Domain 2 presents EIL exceedances for zinc and TRH, and Domain 12 confirms ecological exceedances for zinc and TRH F2 in fill soils, with anomalous selenium and uranium enrichment at a targeted haul road location. Human health criteria exceedances have not been detected during previous investigations.

A number of data gaps exists within this sub-precinct including Domains 5 and 14 which have not been subject to intrusive investigation, and Domain 1 where building footprints and car parks remain uninvestigated and no information exists on removal of the Domain 1 fuel storage facility. Domain 2 lacks intrusive investigation data. Domain 12 contains uncharacterised selenium and uranium enrichment which is considered a lesser constraint.

Based on available data, known or potential contamination is unlikely to preclude conservation rezoning. The absence of human health exceedances and limited ecological findings confirm broad compatibility with conservation land use. However, further investigation is needed to address identified data gaps, particularly for Domain 1 which represents the highest constraint for this sub-precinct. Additional investigation will provide defensible evidence and ensure contamination does not constrain conservation objectives.

5.6.3 Sub precinct 6c

Exceedances of ecological investigation level criteria have been confirmed within domains associated with this sub-precinct. Domain 12 contains ecological criteria exceedances for zinc and TRH in fill materials, with anomalous selenium and uranium enrichment at a targeted location. Human health criteria exceedances have not been detected at Domain 12 during previous investigations. Domain 14 has not been subject to intrusive investigation and as such, potential exceedances within this domain cannot be determined.

Domain 12 lacks surface water assessment, and selenium and uranium enrichment largely remains uncharacterised. Domain 14 represents the most significant data gap as no intrusive investigation has been undertaken across any media. The absence of investigation data in Domain 14 and lack of detailed ecological risk assessment for Domain 12 prevent confident verification of land suitability.

Based on available data, known or potential contamination is unlikely to preclude conservation rezoning. Conservation use is considered compatible with the current contamination profile, provided targeted intrusive investigation of soil and surface water is undertaken in this sub-precinct. Additional investigation and future management controls will provide defensible evidence for any future use.

5.6.4 Sub precinct 6d

Exceedances of investigation levels have been confirmed across multiple domains associated with this sub-precinct. Domain 9 exhibits confirmed EIL exceedances for TRH and zinc in soils, with fibrous asbestos confirmed near former infrastructure. Domain 10 includes ecological exceedances for heavy metals, TRH, and PAHs in maturation pond sediments, with PFOS exceeding the 99% species protection criterion in surface water. Asbestos was confirmed in surface soils near demolished buildings in Domain 11. However, exceedances of human health criteria have been recorded across any domain within this sub-precinct.

Data gaps limit a detailed understanding of the contamination profile for this sub-precinct. Domain 9 presents incomplete wastewater cell assessment, absent groundwater data, and incomplete asbestos characterisation. Domain 10 presents incomplete maturation pond sediment characterisation, limited groundwater data, and asbestos pipework extent undefined. Domain 11 presents uninvestigated building footprints, absent groundwater and surface water data, and no methane assessment. No intrusive investigations have been undertaken for Domain 14.

Based on available data and information, known or potential contamination is unlikely to preclude conservation rezoning of Conservation Area 6d. However, targeted investigation is required within this sub-precinct, particularly for sediment and surface water bodies that occupy the northern portion of this sub-precinct and have not been subject to detailed assessment. Other potential contamination sources associated with Domains 9, 10 and 11 are considered less likely to restrict the capacity of this sub-precinct to be rezoned for conservation on the basis that many are located external to the zoning boundary (as shown in **Figure 7, Appendix A**). Notwithstanding, targeted site inspections and sediment and surface water assessments will enable this area to support conservation land use while protecting environmental values and Lake Macquarie.

5.6.5 Sub precinct 6e

Exceedances of ecological investigation levels have been confirmed across multiple contaminants and domains associated with this sub-precinct. It is noted that this sub-precinct intersects portions of Domain 14 and to a lesser extent Domain 7 (as shown in **Figure 7, Appendix A**). No intrusive investigations have been undertaken within these domains, and consequently no confirmed exceedances of investigation levels have been detected in soil, groundwater, surface water, or sediment. Data gaps therefore remain within this sub-precinct that constrain the ability to confirm land suitability with confidence.

Domain 14 is unlikely to preclude conservation zoning for this sub-precinct. However, Domain 7 represents a material constraint that requires further assessment. Targeted investigations of soil, sediment, surface water, and ground gas is required to resolve remaining uncertainties and confirm land use suitability for future use.

5.6.6 Sub precinct 6f

Exceedances of ecological investigation levels have been confirmed within Domain 12 associated with this sub-precinct. Concentrations of zinc and TRH were reported above adopted ecological screening criteria in fill materials associated with the main haul road. Anomalous enrichment of selenium and uranium was also identified in soils used to construct the haul road associated with Domain 12. No exceedances of human health investigation levels have been detected in soil during previous investigations completed at Domain 12.

Domain 12 lacks surface water assessment and selenium and uranium enrichment largely remain uncharacterised. Domain 14 represents the most significant data gap as no intrusive investigations have been undertaken across any media. However, based on available data, known or potential contamination is unlikely to preclude conservation rezoning as it primarily consists of established bushland. Conservation use is considered compatible with the current contamination profile, provided targeted intrusive investigations of soil and surface water is undertaken in this sub-precinct. Additional investigation and future management controls will provide defensible evidence for the proposed future use.

5.7 Interfaces with other constraints

Management of identified COPCs across the Precinct and Teralba Southgate does not operate in isolation, it interacts with a number of other environmental and engineering constraints that must be considered in an integrated manner during Master Plan planning, rezoning, and future DA-stage assessments. The following section summarises other environmental constraints that require further assessment or consideration at the Precinct and Teralba Southgate.

5.7.1 Waterways and Wetlands

The Precinct and Teralba Southgate is surrounded by sensitive water bodies including Cockle Creek, Lake Macquarie, and associated wetlands. Several domains contain contaminants, including heavy metals, hydrocarbons, nutrients, and PFAS, that may represent a risk of migration to these receiving environments via surface water runoff and groundwater discharge pathways, including Domains 4, 6, 9, 10, and 13. Surface water and groundwater investigations must be undertaken at the DA stage for proposals adjacent to or within the waterway and wetland interface zones, with results assessed against relevant environmental legislation and guidance including (but not limited to) the Australian and New Zealand Guidelines for Fresh and Marine Water Quality (ANZG 2018). Where PFAS is identified as a COPC, assessment must be undertaken in accordance with the PFAS NEMP 3.1. Contamination pathways to these receptors must be identified and controlled during earthworks and development consistent the performance criteria outlined in **Section 6.2**.

5.7.2 Acid Sulfate Soils

As shown in Figure 3 (**Appendix A**), discrete occurrences of potential acid sulfate soils (PASS) have been identified across multiple sub-precincts, including Cockle Creek (sub-precinct 1), Killingworth (sub-precinct 2), Barnsley West (sub-precinct 5), and environmentally constrained sub-precincts 6b, 6c, and 6d. Disturbance of ASS during earthworks or dewatering activities can generate acid leachate that mobilises heavy metals into groundwater and surface water, presenting a risk to receiving waterways and wetlands. While PASS may underlie some areas, it is not expected that this will likely be disturbed based on the current zoning plan. However, where PASS may be disturbed, an Acid Sulfate Soil Management Plan (ASSMP) must be prepared and implemented prior to earthworks in relevant sub-precincts, consistent with the performance criteria outlined in **Section 6.2**. The ASS risk must be considered in the context of geotechnical and dewatering planning at the DA stage to ensure integrated management outcomes.

5.7.3 Salinity

As shown in Figure 4 (**Appendix A**), portions of the Precinct and Teralba Southgate are classified as moderate to very high salinity hazard zones, associated with low-lying areas, disturbed terrain, and the Cockle Creek floodplain interface. Elevated salt concentrations reflect historical mining operations, evaporative concentration in surface water structures, and estuarine influence from Lake Macquarie. Salinity hazard presents a material constraint to sensitive land use rezoning and may trigger additional assessment requirements at the development application stage.

Within environmentally constrained areas (6a–6f), elevated salinity may preclude low-salinity native vegetation establishment, requiring soil remediation, salt-tolerant species selection, or hydrological management. In areas where structures are proposed, elevated salinity increases corrosivity risk to buried infrastructure and requires assessment of foundation stability where shallow groundwater and high evaporative demand may induce heave or differential settlement. Dewatering and excavation require management to prevent salt-laden leachate mobilisation.

Salinity risk should be integrated into site-wide water cycle management and contamination control planning. Detailed salinity assessment, including soil and groundwater profiling, should be undertaken for Teralba Southgate and all sub-precincts proposed for sensitive land uses prior to development approval. Salinity management measures including hydrological controls, soil amendment, or engineered containment should be embedded in Environmental Management Plans (EMPs) and validated through post-construction monitoring to confirm effectiveness and ongoing compliance with land use suitability criteria.

5.7.4 Tailings Storage Facility

The Tailings Storage Facility (TSF) in Domain 7 represents a discrete and material constraint that operates across multiple environmental and engineering disciplines, requiring integrated management during Master Plan planning and future development application assessments. Unlike contamination in other domains, the TSF presents a fundamental constraint to rezoning and redevelopment that extends beyond standard remediation pathways, necessitating coordinated consideration with water management, geotechnical stability, dam safety, and regulatory compliance frameworks.

As outlined in **Section 3-4**, the TSF is currently being remediated to ensure the final landform is stable and environmentally sound, however has not been subject to detailed investigation to inform potential contamination risks. TSFs are known to contain tailings materials with unknown geochemical characteristics including potential acid-forming materials (PAF), spontaneous combustion potential, and salinity risks. Leachate generation and acid mine drainage (AMD) from tailings represent primary contamination pathways to downgradient groundwater and surface water environments. These pathways may directly interfere and constrain rezoning proposals or development applications, requiring detailed assessment by suitably qualified environmental consultants. On this basis, Aurecon recommends that further contamination investigations of soil, sediment, groundwater and surface water at and downgradient of the TSF be undertaken prior to rezoning and future development applications.

5.7.5 Earthworks and Spoil Management

Significant bulk earthworks are anticipated across the Precinct as part of future development. If offsite disposal of material is required during future construction programs, subject material must be classified in accordance with the *NSW EPA Waste Classification Guidelines, Part 1: Classifying Waste* (2014) and PFAS Addendum (2016). Toxicity Characteristic Leaching Procedure (TCLP) analysis may also be required for soils to confirm final waste categories. Where soils meet the criteria for Virgin Excavated Natural Material (VENM) or Excavated Natural Material (ENM), reuse as engineered fill offsite should be considered for relevant materials under the Resource Recovery Order or Resource Recovery Exemption (RRO/RRE) to minimise disposal costs and waste being sent to landfill. A Spoil Management Plan must be prepared prior to bulk earthworks commencing in accordance with the performance criteria outlined in **Section 6.2**.

5.7.6 Water Reuse Concepts

Mine water retained in final void of Domain 4 may present opportunities for reuse in construction activities, irrigation practices, and/or recreational purposes across the Precinct, subject to water quality assessment and appropriate approvals. The potential for AMD and elevated contaminants in mine water must be assessed before any reuse pathway is confirmed, with results assessed against relevant guidance including (but not limited to) the ANZG 2018, Australian Guidelines for Water Recycling, and Guidelines for Managing Risks in Recreational Water (NHMRC 2008). Water reuse concepts must be developed in an integrated manner with contamination management, drainage design, and the Precinct's water cycle strategy to ensure that water quality risks are appropriately managed.

5.7.7 Ground Gas

Underground workings across the Precinct and Teralba Southgate present a potential methane and ground gas risk, with detections recorded during previous investigations. Ground gas characterisation across the Precinct and Teralba Southgate is limited by the scope and frequency of previous investigations. Elevated methane has been detected in mine shafts in Domain 11 and during drilling in Domain 13 (Teralba Southgate) where the 5% Lower Explosive Limit (LEL) was exceeded. While subsequent assessments at these domains did not detect elevated ground gas concentrations, Aurecon considers that these investigations were limited in scope. Given the ubiquitous nature of methane in underground environments, particularly within abandoned workings and associated underground structures, further assessment is required to establish baseline conditions and define risk zones prior to development.

Ground gasses have the potential to accumulate within buildings, basements, and service trenches creating potential risks of vapour intrusion, explosive atmospheres, and asphyxiation in future structures or confined spaces. Ground gas risk may also represent an explosive or exposure risk if encountered during future earthworks undertaken at the Precinct or Teralba Southgate. Early assessment and targeted mitigation are therefore essential for future development.

Where ground gas identified as a potential constraint, a hazardous ground gas assessment must be undertaken per the NSW EPA Assessment and Management of Hazardous Ground Gas Guidelines (2020), focusing on characterising ground gas at relevant depths and monitoring within buildings, service pits, drains, and enclosed spaces prior to occupation. Results must be assessed against relevant criteria to determine land use compatibility. Residual risks must be managed through engineering controls (e.g. ventilation, gas-resistant membranes, sub-slab depressurisation) consistent with **Section 6.2** performance criteria, integrated with site-wide contamination and geotechnical planning.

6 Master Plan objectives and performance criteria

The following objectives and performance criteria have been developed to guide the management of contamination across the Precinct through future planning approvals and development stages. They are intended to be read in conjunction **Sections 4** and **5** of this report. As outlined in **Section 1.1**, Teralba Southgate has been removed from the scope of the Master Plan and is therefore excluded from this section.

6.1 Contamination Master Plan Objectives

Master plan objectives establish the framework for matching contamination risk profiles to proposed land uses. **Table 6-1** outlines the master plan objectives and rationale for the Precinct.

Table 6-1 Master Plan Objectives

Objective ID	Objective	Rationale
O1	Enable employment-led reuse while managing contamination risk through a staged, risk-based assessment and remediation pathway.	The Precinct-scale evidence base confirms that contamination is generally manageable for industrial/commercial uses; however, a staged pathway provides regulatory certainty without front-loading investigation costs.
O2	Avoid placing higher sensitivity uses in areas where contamination uncertainty or remediation burden is likely to be high.	Certain domains contain data gaps or confirmed contamination that would require extensive investigation and remediation to support more sensitive land uses (conservation, residential or recreational).
O3	Protect waterways and ecological assets by managing soil and groundwater disturbance risk.	Cockle Creek, Lake Macquarie, and surrounding associated wetlands are sensitive ecological receptors present near the Precinct. Contamination pathways to these receptors must be identified and controlled during earthworks and development to ensure their ongoing protection
O4	Ensure that contamination management is integrated with earthworks planning and water cycle management across the Precinct.	Coal mine sites involve significant bulk earthworks and water management challenges that interact directly with contamination risk. Failure to integrate contamination management with earthworks and water cycle planning at the Master Plan level risks unexpected contamination encounters during construction phase. Early integration ensures that soil management plans, waste classification requirements, and water quality controls are embedded in project design from the outset.
O5	Ensure remediation and validation outcomes are completed to a standard appropriate for the proposed land use, with ongoing monitoring maintained where residual risks remain.	Validation sampling and reporting, as well as monitoring programs must be submitted as part of future development applications where remediation has been undertaken. Per LMCC requirements, investigation, remediation and validation works should be conducted by a suitably qualified environmental consultant, under the direction of an accredited CEnvP-SC or equivalent. In some instances, an accredited NSW EPA Auditor may be required to endorse that remedial and validation works have been completed in accordance with the relevant legislation and guidelines and that potentially unacceptable risks to human health or the environment have been mitigated.

Objective ID	Objective	Rationale
O6	Manage risks associated with underground mine workings, including hazardous ground gas, through targeted assessment prior to construction of enclosed structures.	Underground workings across the Precinct present a potential methane and ground gas risk of vapour intrusion, creating an explosive environment and/or risk of asphyxiation in future above ground structures or below ground confined spaces such as basements or service trenches. Early assessment ensures that appropriate mitigation measures are incorporated into building design and construction methodology.

6.2 Contamination Performance Criteria

The performance criteria below define the minimum evidence and management requirements for future development proposals and apply to all land use zones and development applications across the Precinct. **Table 6-2** outlines the performance criteria for the Precinct in the context of the proposed future land uses.

Table 6-2 Summary of Performance Criteria

Performance criteria ID	Performance Criterion
PC1 - Land Use Suitability	Development applications must be accompanied by a contamination assessment that demonstrates the subject site is suitable or can be made suitable for the proposed land use following remediation as per the requirements of the <i>State Environmental Planning Policy (Resilience and Hazards) 2021</i>. Future land use suitability assessments must be undertaken in accordance with relevant legislation and guidance including (but not limited to) the <i>Contaminated Land Management Act</i> (CLM Act), <i>NEPM 2013</i>, <i>PFAS National Environmental Plan</i> (NEMP 3.1), <i>Assessment and Management of Hazardous Ground Gas</i> (2020), and <i>Guidelines for the NSW Site Auditor Scheme 3rd edition</i> (2017) and other relevant legislation and guidelines as required.
PC2 - Water Quality Protection	Development must not result in the degradation of surface water or groundwater quality and must demonstrate that contamination pathways to receiving environments are identified and appropriately managed. Future surface or ground water investigations will need to be undertaken in accordance relevant guidance noted above and (but not limited to) the Australian and New Zealand Guidelines for Fresh and Marine Water Quality (ANZG 2018), ANZECC & ARMCANZ 2000 and the NEMP 3.1.
PC3 - Validation and Ongoing Monitoring	Where remediation or management measures are implemented, they must be completed in accordance with the Hazards and Resilience SEPP (2021) and the National Remediation Framework. A site-specific Remediation Action Plan must be developed to be considerate of the proposed land use/s and design. The RAP must outline validation requirements that must be completed and reported prior to subdivision or occupation. Ongoing monitoring must be maintained where residual risks remain which may require a Long-Term Environmental Management Plan (LTEMP). Validation and monitoring activities must be undertaken in accordance with the legislation and guidance referenced in PC1 above.
PC4 -Hazardous Ground Gas Assessment	Hazardous ground gasses must be assessed and remediated/managed in accordance with relevant guidance including (but not limited to) the <i>NSW EPA Guidelines on the Assessment and Management of Hazardous Ground Gases 2020</i>.
PC5 - Spoil Management	On a site or project specific basis, a Spoil Management Plan (SMP) is prepared and approved prior to bulk earthworks commencing, identifying volumes, characterisation findings, proposed reuse/disposal pathways, and contingency measures for unexpected contamination.

Performance criteria ID	Performance Criterion
PC6 - Waste Characterisation and Management	All waste management activities must comply with the <i>Protection of the Environment Operations Act 1997</i> (POEO Act), <i>Waste Avoidance and Resource Recovery Act 2001</i> (WARR Act), and relevant EPA licencing conditions. Any waste proposed for offsite disposal must be characterise in accordance with the <i>NSW EPA Waste Classification Guidelines</i> (2014) and PFAS addendum (2016), Resource Recovery Exemption/Resource Recovery Order (RRE/RRO), and guidance specific to assessing virgin excavated natural material (VENM) All waste must be disposed to a licenced facility or otherwise reused offsite as VENM or under a RRO/RRE. The appropriate number of soil samples to characterise waste should be collected in accordance with the <i>NSW EPA Sampling Design Guidelines 2022</i>.
PC7 - Unexpected Finds Protocol	An Unexpected Finds Protocol must be established and implemented prior to and during any future demolition, excavation, or civil earthworks.

7 Investigation and remediation triggers

The investigation and remediation triggers presented in this section define the circumstances under which further contamination assessment, remediation, and validation are required across the Precinct and Teralba Southgate. The triggers are staged and risk-based, with the nature and extent of investigation required determined by land use sensitivity, disturbance footprint, and the contamination risk profile of the domain.

Table 7-1 summarises the investigation and remediation triggers applicable to future development proposals across the Precinct and Teralba Southgate.

Table 7-1 Investigation and Remediation Triggers

Investigation Stage	Trigger Condition	Applicable sub precincts (indicative)	Key Outputs
Preliminary Site Investigation (PSI)	Any development proposal or rezoning proposal	All sub precincts	AEC confirmation, updated CSM, data gap identification, recommendation for DSI or no further action.
Sampling Analysis and Quality Plan (SAQP)	PSI or previous investigations indicates that potential contamination sources may be at the subject site	All sub precincts and Teralba Southgate.	Sampling strategy for AECs including data quality objectives and indicators
Detailed Site Investigation (DSI)	PSI or previous investigations identifies potential contamination risks or data gaps insufficient for a rezoning or land use decision.	All sub precincts and Teralba Southgate.	Nature and extent of contamination characterised, risk assessment, waste classification, RAP trigger assessment.
Tier 2 or 3 risk assessment	Concentrations of COPCs detected above Tier 1 screening values that represent a potential risk to human health or the environment and are proposed to be retained on site.	Dependant on DSI findings	Site-specific risk assessment, derivation of site-specific criteria, recommendation for RAP or no further action.
Remedial Action Plan (RAP)	DSI or risk assessment identifies potentially unacceptable risk for the current or proposed land use.	As identified by DSI	Remediation strategy, validation criteria, regulatory notification requirements.
Validation Report	Completion of remediation works; prior to subdivision, occupation, or change of use. A NSW EPA accredited Site Audit may be required where more sensitive land uses are proposed.	All remediated areas	Confirmation of suitability for proposed use, site audit statement pathway where required.

Investigation Stage	Trigger Condition	Applicable sub precincts (indicative)	Key Outputs
Unexpected Finds Protocol (UFP)	All earthworks, demolition, or civil works across sub precincts and Teralba Southgate, regardless of prior investigation status.	All sub precincts and Teralba Southgate.	Contingency management of unforeseen contamination, ACM, or buried infrastructure.

8 Post exhibition management framework and approvals pathway

This section describes how contaminated land will be managed through future planning and development approvals, consistent with the master plan objectives and performance criteria outlined in **Sections 6 and 7**. The management framework is structured around the post-exhibition approvals pathway and identifies both the obligations and opportunities associated with contamination management across the Precinct and Teralba Southgate.

8.1 Development application

Development applications for land within the Precinct and Teralba Southgate must be supported by site specific contamination assessments that confirm site suitability for the proposed use, considerate of the engineering design, and identify any remediation or management measures required. The nature and extent of assessment required at the DA stage will be determined by the land use sensitivity, disturbance footprint, and contamination risk profile of identified contamination sources, consistent with performance criteria 1 (PC1) in **Section 6.2**.

Through this approach, contamination risks will be addressed progressively and proportionately, supporting the safe and sustainable transformation of the subject site(s) in line with the proposed vision for the Precinct and Teralba Southgate. The requirement and triggers for contamination assessment during future DA stages are outlined in **Section 7**. All contamination assessments must be prepared by a suitably qualified contaminated land consultant, consistent with LMCCs requirements for development on contaminated land and must comply with the *NSW EPA Guidelines for Consultants Reporting on Contaminated Sites* (2020). Where reports are submitted to the NSW EPA or otherwise required by the consent authority, assessments must be prepared or reviewed by a consultant certified under an EPA-recognised scheme (CEnvP-SC) or equivalent.

8.2 Regulatory notifications and duty to report

Under Section 60 of the *Contaminated Land Management Act* 1997 (CLM Act), a person who becomes aware of contamination of land in NSW must notify the NSW EPA if the contamination is of a type, nature, or extent prescribed by the regulations.

As outlined in this report, several domains contain elevated concentrations of COPCs, however available data does not confirm contamination at concentrations or extents sufficient to trigger a notification obligation under Section 60 of the CLM Act at this time. Aurecon notes that this conclusion is limited by the data gaps identified in **Section 3.5**, and a number of domains remain largely unassessed or partially investigated. The duty to report must therefore be reviewed and updated at each subsequent DA-stage investigation as additional intrusive data becomes available. Where future investigations identify contamination that may trigger reporting obligations, the relevant landowner or proponent must notify the NSW EPA in accordance with the CLM Act and the *NSW EPA Guidelines on the Duty to Report Contamination* under the CLM Act 1997 (EPA 2015).

8.3 Opportunities

The Precinct and Teralba Southgate present a number of opportunities to manage contaminated and potentially contaminated materials in a cost-effective and environmentally responsible manner. Identifying and pursuing these opportunities at future DA stages can significantly reduce off-site disposal costs and minimise the environmental footprint of future development. The following opportunities are identified for consideration in future development planning:

- **Existing data and information** – the contamination investigations completed to date provide a substantial and layered evidence base across multiple domains that can be directly built upon for any required DSIs. In areas that have already been characterised to an appropriate level, the scope of new intrusive investigation can be minimised, reducing both cost and program risk for future development proposals
- **On-site bioremediation** – hydrocarbon impacted soils may be suitable for treatment via an on-site bioremediation pad, where excavated material is treated in controlled conditions before being validated for reuse or disposal. This approach can significantly reduce off-site disposal volumes and associated costs for domains with confirmed hydrocarbon contamination
- **Onsite reuse as engineered fill** – excavated soils that have been subject to an appropriate level of investigation and meets NEPM land use criteria and does not contain indicators of contamination can be reused onsite
- **Offsite reuse as engineered fill** – excavated soils that meet the criteria for VENM or ENM may be reused offsite as engineered fill, subject to further classification and validation. RROs/RREs should be considered where applicable for aggregates, slag or other eligible materials
- **Waste characterisation and soil leachate analysis** – TCLP analysis may be undertaken on contaminated soils to assess their potential for leaching and determine appropriate waste streams. This is particularly relevant for domains with PFAS, hydrocarbon or heavy metal contamination where soils require disposal to a licensed waste facility
- **Mine water reuse** – treated mine water may present an opportunity for reuse in construction activities or irrigation across the Precinct and Teralba Southgate, subject to water quality assessment and appropriate approvals. This should be considered in the context of integrated water cycle planning for the Precinct and Teralba Southgate.

9 Conclusions and recommendations

9.1 Conclusions

This assessment included a review of publicly available information and previous investigations completed progressively at the Precinct and Teralba Southgate since 2005 across several Domains or Areas of Environmental Concern. The findings of this assessment confirm that soil, groundwater, surface water quality are generally compatible with industrial and commercial land uses through implementation of a staged, risk-based assessment and remediation pathway. However, more sensitive land uses including residential, conservation, and recreational uses require targeted investigation and risk assessment to confirm suitability and establish appropriate remediation and management controls.

Available data and information demonstrate that exceedances of human health investigation levels for commercial/industrial land use were not identified during previous investigations. However, ecological investigation level exceedances were reported in soil and sediment, reflecting the legacy of coal preparation operations. These ecological impacts require management to protect sensitive receptors including Cockle Creek and Lake Macquarie, but do not preclude employment-led development provided appropriate assessment and management controls are implemented. Contamination sources including asbestos in fill materials, PFAS in soil and groundwater, and hazardous ground gases have been confirmed across multiple domains and require targeted management consistent with the performance criteria outlined in this assessment.

Significant data gaps constrain the ability to fully characterise the contamination profile across all domains. Five domains have not been subject to intrusive investigations, and multiple domains lack groundwater, surface water, sediment, or soil vapour/ground gas assessment data. These gaps are most material for sensitive land use scenarios including residential, recreation and conservation. The TSF (Domain 7) represents a material constraint requiring integrated assessment with water management, geotechnical stability, and dam safety considerations. Implementation of the staged investigation program will progressively resolve these data gaps through the development application approval process, enabling risk-based decision-making.

The management framework established in this assessment provides a clear pathway for contamination management through future planning and development approvals. Development applications must be supported by site-specific assessments that confirm suitability for the proposed use based on remediation or management measures required. Six performance criteria have been established to guide contamination management:

- Land Use Suitability,
- Water Quality Protection,
- Validation and Ongoing Monitoring,
- Hazardous Ground Gas Assessment,
- Spoil Management, and
- Waste Characterisation and Management.

Investigation and remediation triggers have also been defined to ensure that the nature and extent of assessment is proportionate to land use sensitivity, disturbance footprint, and contamination risk profile.

The Precinct and Teralba Southgate present opportunities to manage contaminated materials cost-effectively, including on-site bioremediation of hydrocarbon-impacted soils, onsite or offsite reuse as engineered fill, and mine water reuse in construction activities. Implementation of the management framework and performance criteria outlined in this assessment, combined with progressive investigation and remediation through the development application approval process, will enable the Precinct and Teralba Southgate to support sustainable post-mining land use while protecting human health and environmental values in Cockle Creek and Lake Macquarie. The staged, risk-based approach provides regulatory certainty and defensible evidence for rezoning decisions while enabling cost-effective contamination management aligned with the proposed employment-led vision for the Precinct and Teralba Southgate.

9.2 Recommendations

Based on the conclusions provided above, and subject to the limitations outlined in **Section 11**, Aurecon presents the following recommendations:

- Implement a risk-based, staged investigation program for all future development applications, with Detailed Site Investigations (DSIs) and Tier 2/3 quantitative risk assessments undertaken where Tier 1 screening level concentrations exceedances or data gaps are identified, ensuring land use suitability is confirmed prior to development approval
- Prioritise targeted investigation of Domains 2, 5, 7, 14, and 16 (uninvestigated domains), comprehensive groundwater assessment across all domains, ASS characterisation in low-lying areas, and asbestos delineation sampling to resolve material data gaps constraining sensitive land use suitability determination. Additionally, it is also recommended that further assessment of the environmentally constrained areas be undertaken, primarily targeting the medium to high-risk domains (1, 7, 9, and 11), with lower priority assessment of the low-risk domains to provide broader site coverage where data gaps exist. This is to ensure that COPCs from other domains have not migrated into the environmentally constrained areas
- Undertake integrated assessment of the TSF across water management, geotechnical stability, dam safety, and regulatory compliance frameworks, including geochemical characterisation of tailings materials and assessment of leachate generation and AMD pathways to inform remediation and long-term management strategy
- Undertake targeted surface water and sediment quality assessments adjacent to waterway and wetland interface zones (including recreational areas within West Lake); and assess PFAS-impacted areas in accordance with the PFAS National Environmental Management Plan (NEMP 3.1)
- Undertake comprehensive ground gas assessment in accordance with NSW EPA Assessment and management of hazardous ground gases (EPA 2020), with particular focus on Domains 11 and 13 where methane concentrations were reported. If required, implement ground gas mitigation measures during building design and construction to prevent vapour intrusion risk to future enclosed structures
- Confirm land suitability for more sensitive uses (residential in Teralba, environmentally constrained areas, and recreational uses) through targeted investigation of soil, groundwater, surface water, and ground gas; implement PFAS plume delineation, asbestos delineation, and methane management measures; obtain Site Audit Statements from NSW EPA-accredited Site Auditors where remediation or management measures are implemented
- During construction, implement contamination pathway controls during earthworks and development to prevent migration of identified COPCs to Cockle Creek and Lake Macquarie
- Ensure all contamination assessments comply with NSW EPA Guidelines and are prepared by a suitably qualified contaminated land consultant including any review and update of duty to report obligations under Section 60 of the Contaminated Land Management Act 1997 at each DA-stage investigation and preparation of Long-Term Environmental Management Plans (LTEMPs) where residual contamination risks remain. As per Lake Macquarie City Council's (LMCCs) requirements, an EPA certified consultant must be engaged where reports are submitted to the NSW EPA or where required by the consent authority
- All future works should be conducted by an appropriately qualified environmental consultant under the direction of a CEnvP-SC or equivalent accreditation. An accredited NSW EPA Auditor may be required to endorse the investigation, remediation/management and validation works where more sensitive land uses are proposed or where there are elevated risks to ecological receptors.

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11 Limitations

This report has been written to inform the Master Plan being prepared for the Precinct and Teralba Southgate. Subsurface conditions relevant to future construction work should be assessed by contractors who can make their own interpretation of the factual data provided and perform any additional tests as necessary for their own purposes and determine the suitability of particular techniques and equipment for the conditions.

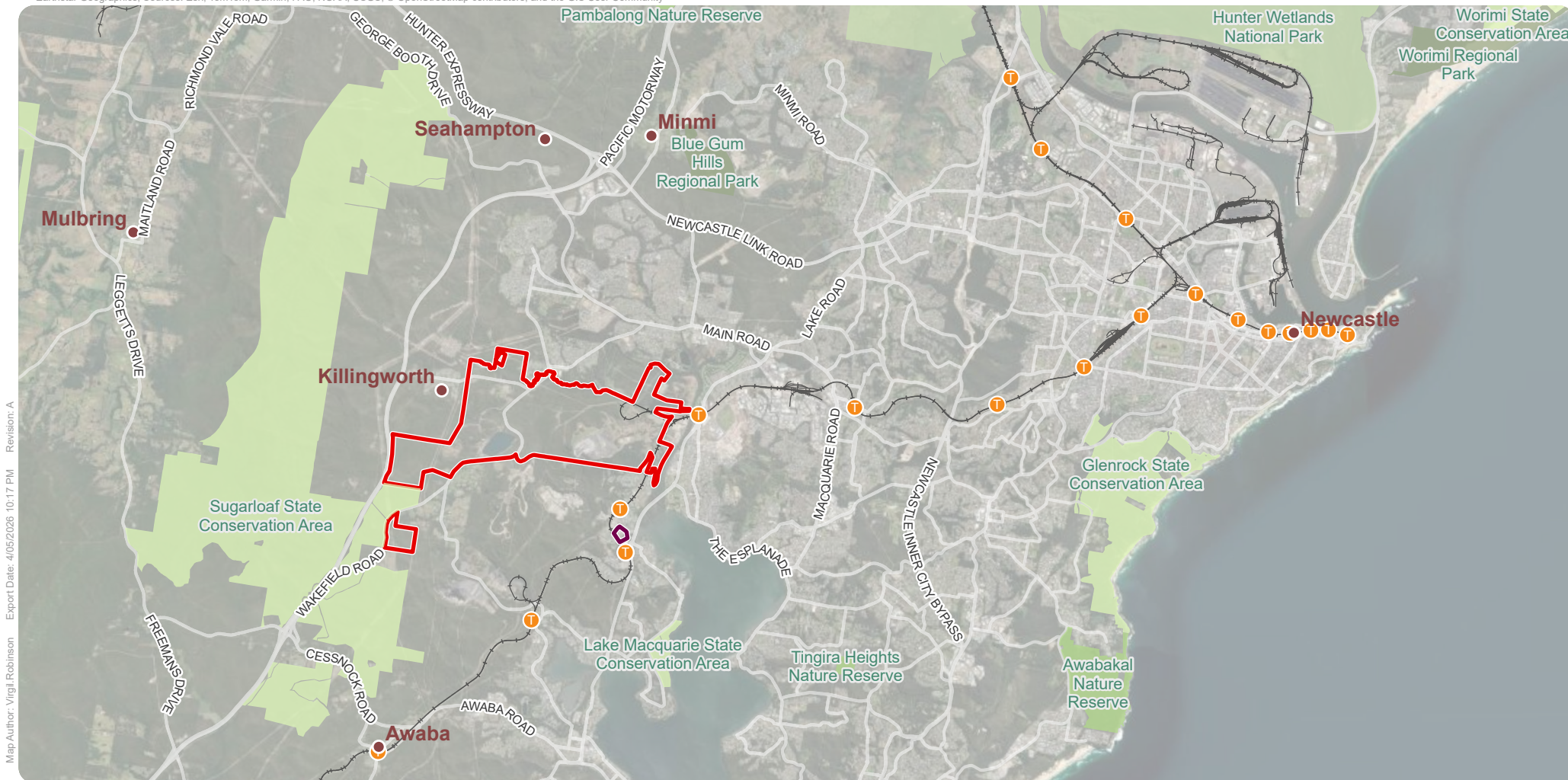
In preparing the report, Aurecon has relied upon publicly available databases, data, surveys, analyses, designs, plans and other information provided by the DPHI, most of which are referred to in the report (the data). Except as otherwise stated in the report, Aurecon has not verified the accuracy or completeness of the data. To the extent that the statements, opinions, facts, information, conclusions and/or recommendations in the report (conclusions) are based in whole or part on the data, those conclusions are contingent upon the accuracy and completeness of the data. Aurecon will not be liable in relation to incorrect conclusions should any data, information or condition be incorrect or have been concealed, withheld, misrepresented or otherwise not fully disclosed to Aurecon.

In accordance with the scope of services, Aurecon has relied upon the data and has not conducted any environmental field testing, sampling or laboratory analyses in the preparation of the report. The conclusions are based upon the data sources included in this report, including field surveys undertaken by Aurecon, and are therefore merely indicative of the environmental condition of the site at the time of preparing the report, including the presence or otherwise of contaminants or emissions.

Within the limitations imposed by the scope of services, the assessment of the site and preparation of this report have been undertaken and performed in a professional manner, in accordance with generally accepted practices and using a degree of skill and care ordinarily exercised by reputable environmental consultants under similar circumstances. No other warranty, expressed or implied, is made.

Appendices

Appendix A – Figures



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| | Precinct boundary | | Railway | | Nature Reserve |
| | Teralba South Gate | | Town | | State Conservation Area |
| | Train station | NPWS Estate | | | Regional Park |
| | | | | | National Park |

Sources: Aurecon, NSW DCCEEW, NSW DCS, Esri Basemap

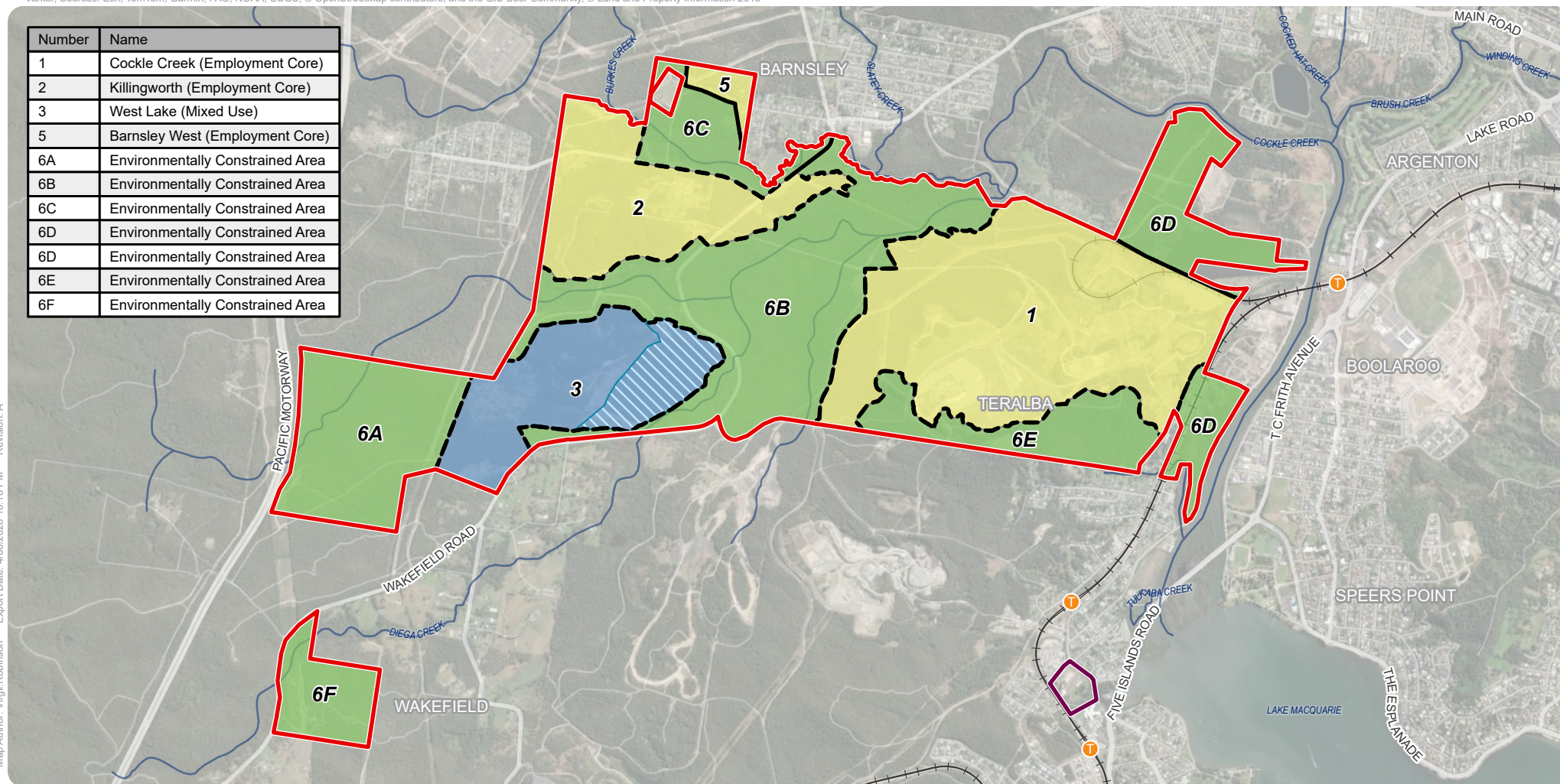


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Number	Name
1	Cockle Creek (Employment Core)
2	Killingworth (Employment Core)
3	West Lake (Mixed Use)
5	Barnsley West (Employment Core)
6A	Environmentally Constrained Area
6B	Environmentally Constrained Area
6C	Environmentally Constrained Area
6D	Environmentally Constrained Area
6D	Environmentally Constrained Area
6E	Environmentally Constrained Area
6F	Environmentally Constrained Area



- Precinct boundary
- Teralba South Gate
- Subprecinct boundary
- T Train station
- + Railway
- ~ Watercourse
- Landuse**
- Environmentally Constrained Area
- Mixed Use
- Recreation Area
- Employment Core

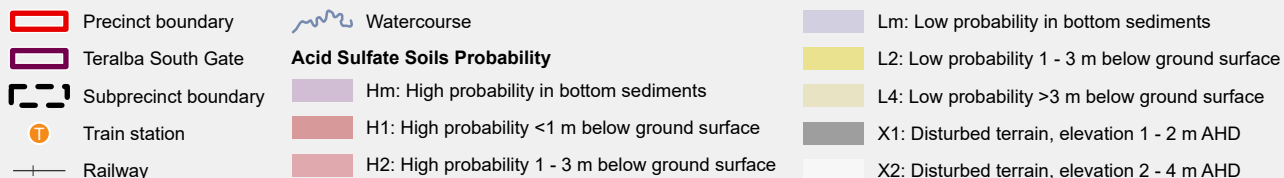
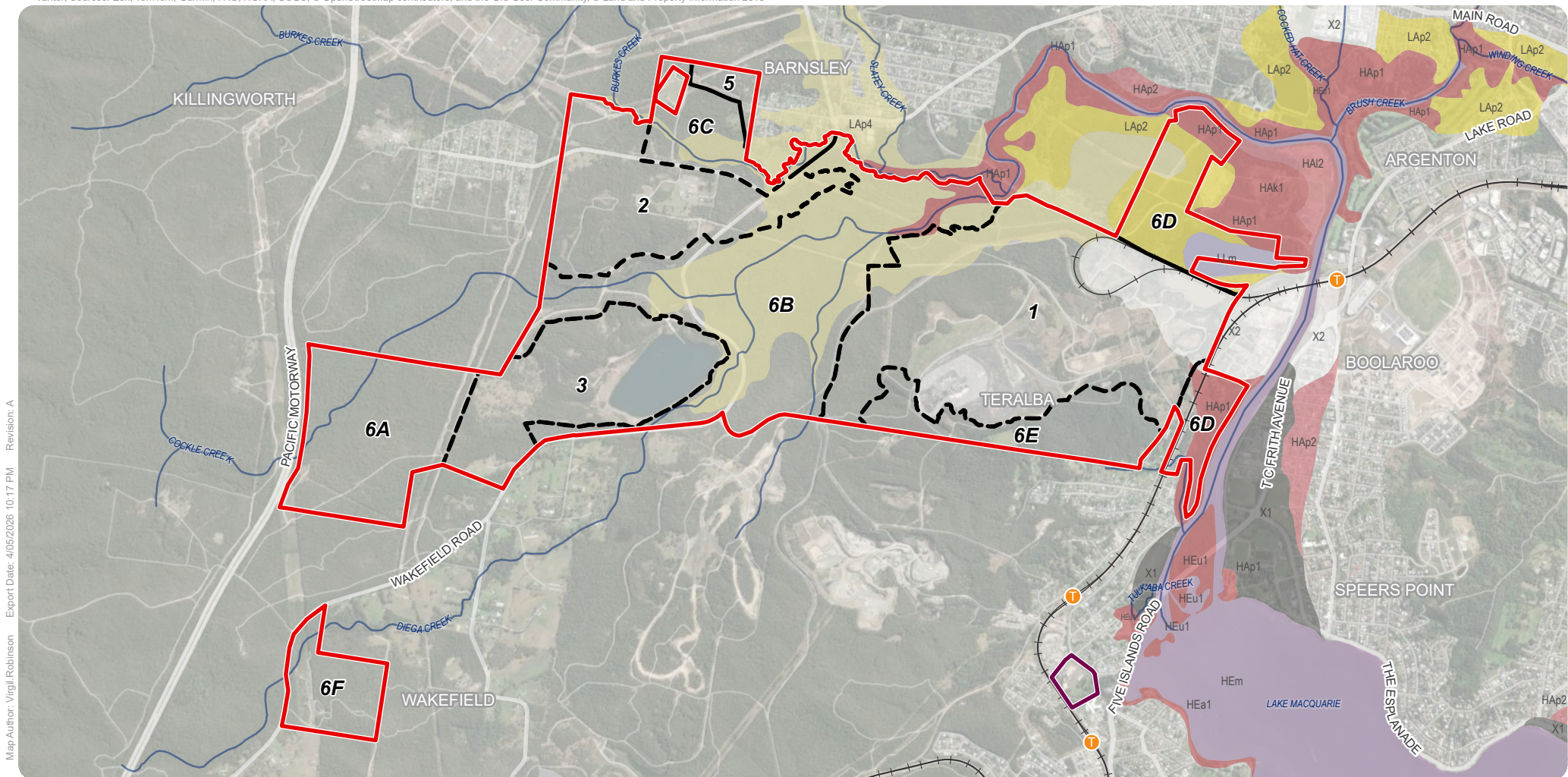
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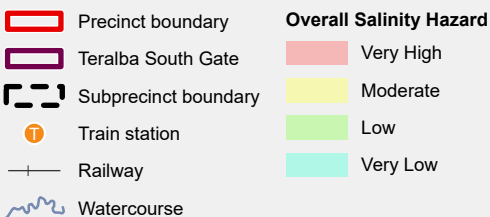
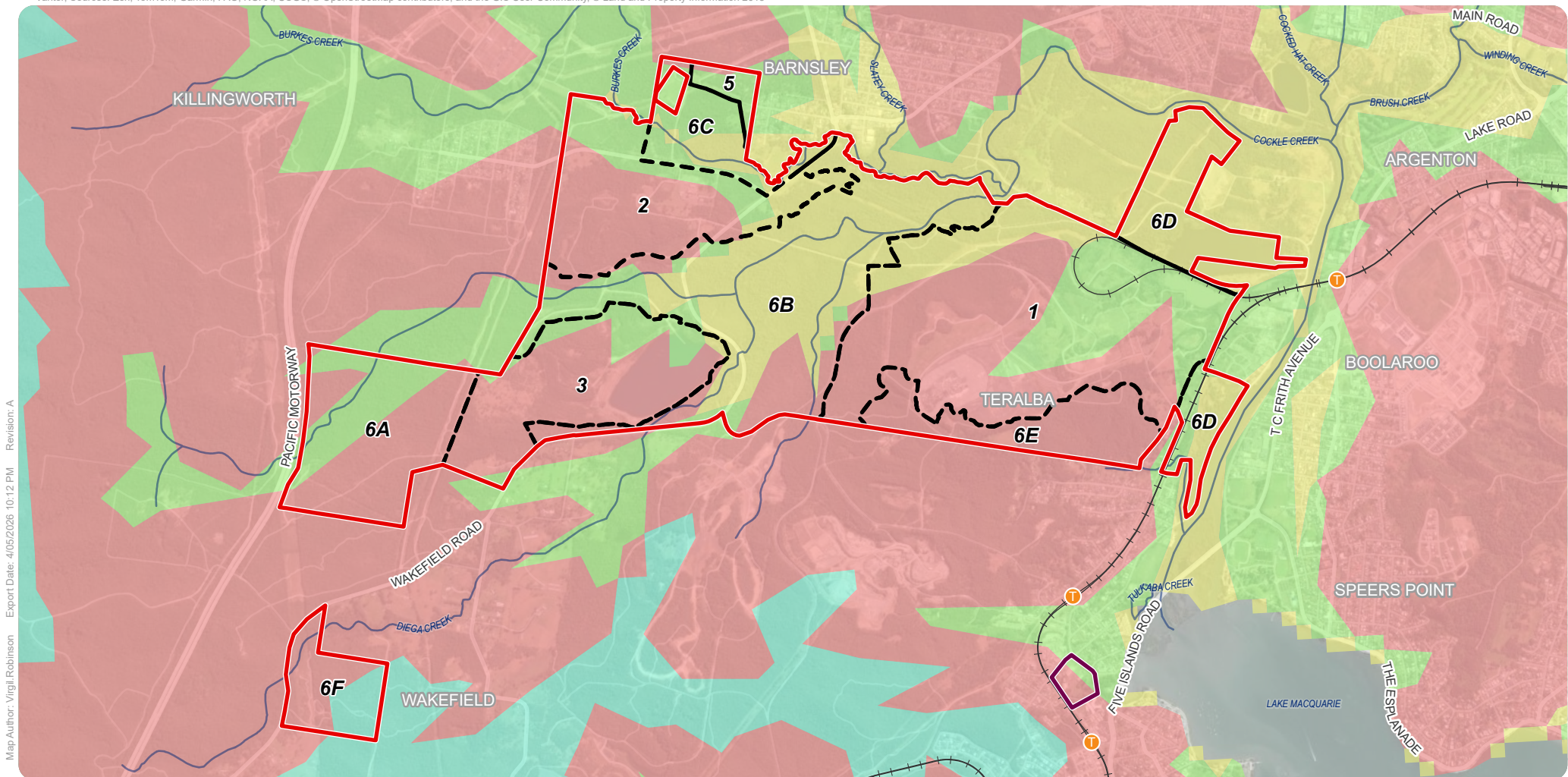
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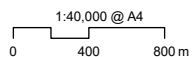
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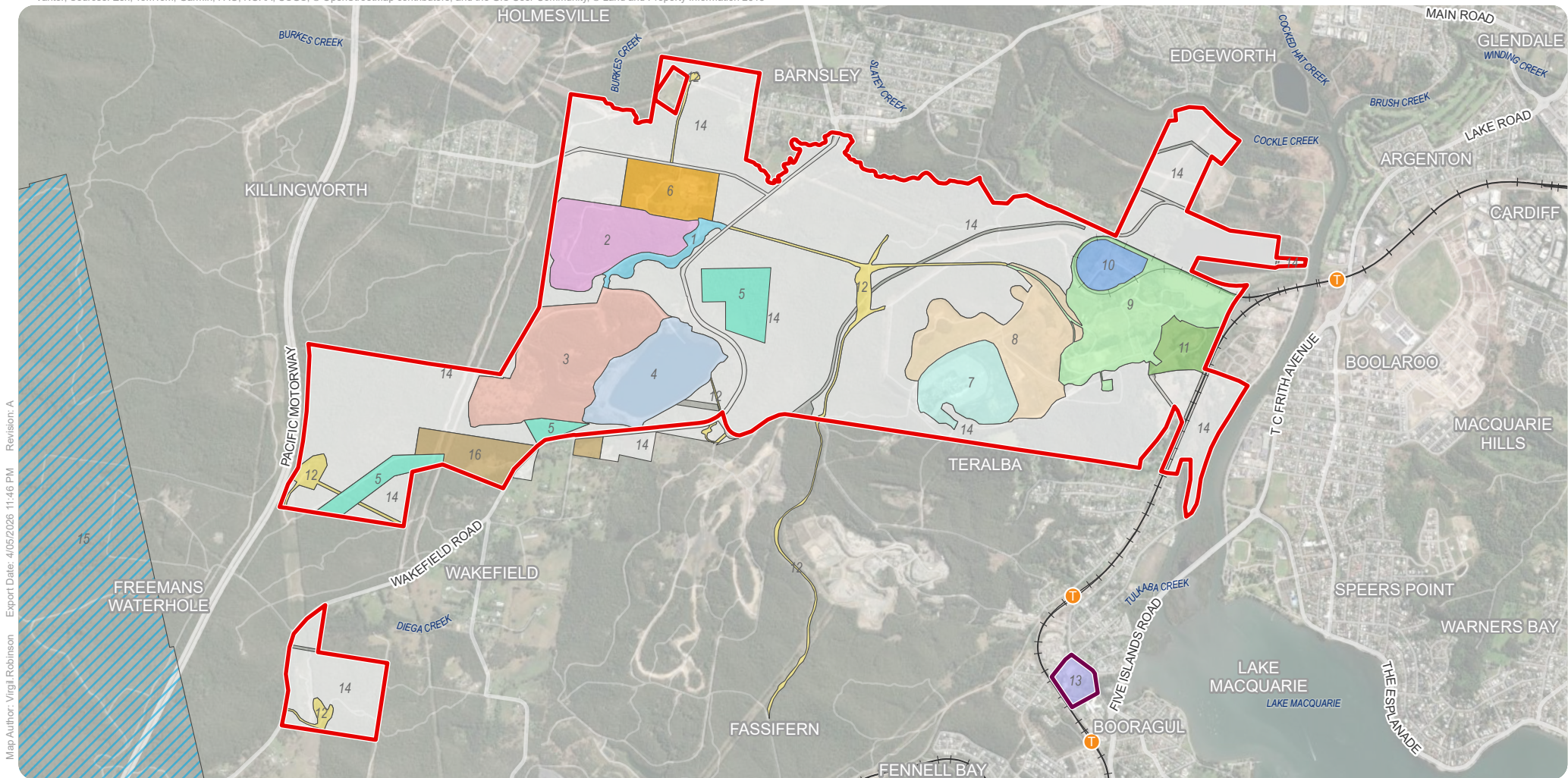


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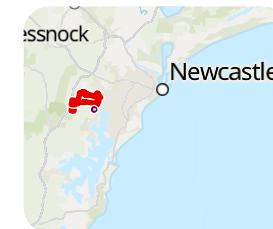
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| Precinct boundary | Domain 2 - North Rehab Area | Domain 8 - Rejects Emplacement Area | Domain 14 - Other Mine Owned Lands |
| Teralba South Gate | Domain 3 - South West Pit | Domain 9 - Surface Facilities | Domain 15 - Project Approval Area |
| Train station | Domain 4 - Proposed Final Void | Domain 10 - Maturation Ponds | Domain 16 - Rural Residential property |
| Railway | Domain 5 - VCA Areas | Domain 11 - Teralba North Gate | |
| Closure Domains (Primary) | Domain 6 - Pit Top Facilities | Domain 12 - Infrastructure | |
| Domain 1 - Admin Workshop Haul Rd | Domain 7 - Tailings Dam | Domain 13 - Teralba South Gate | |

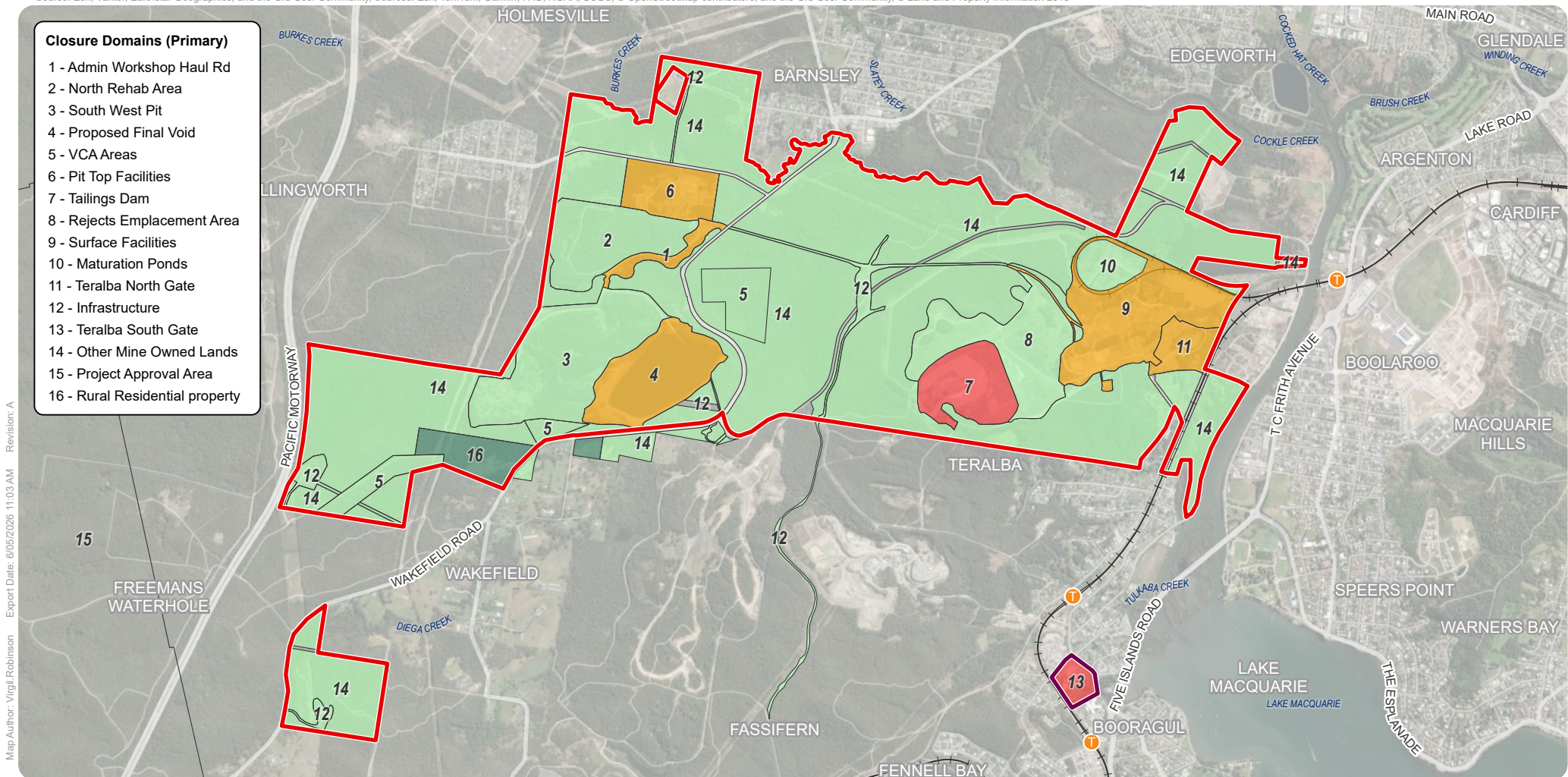
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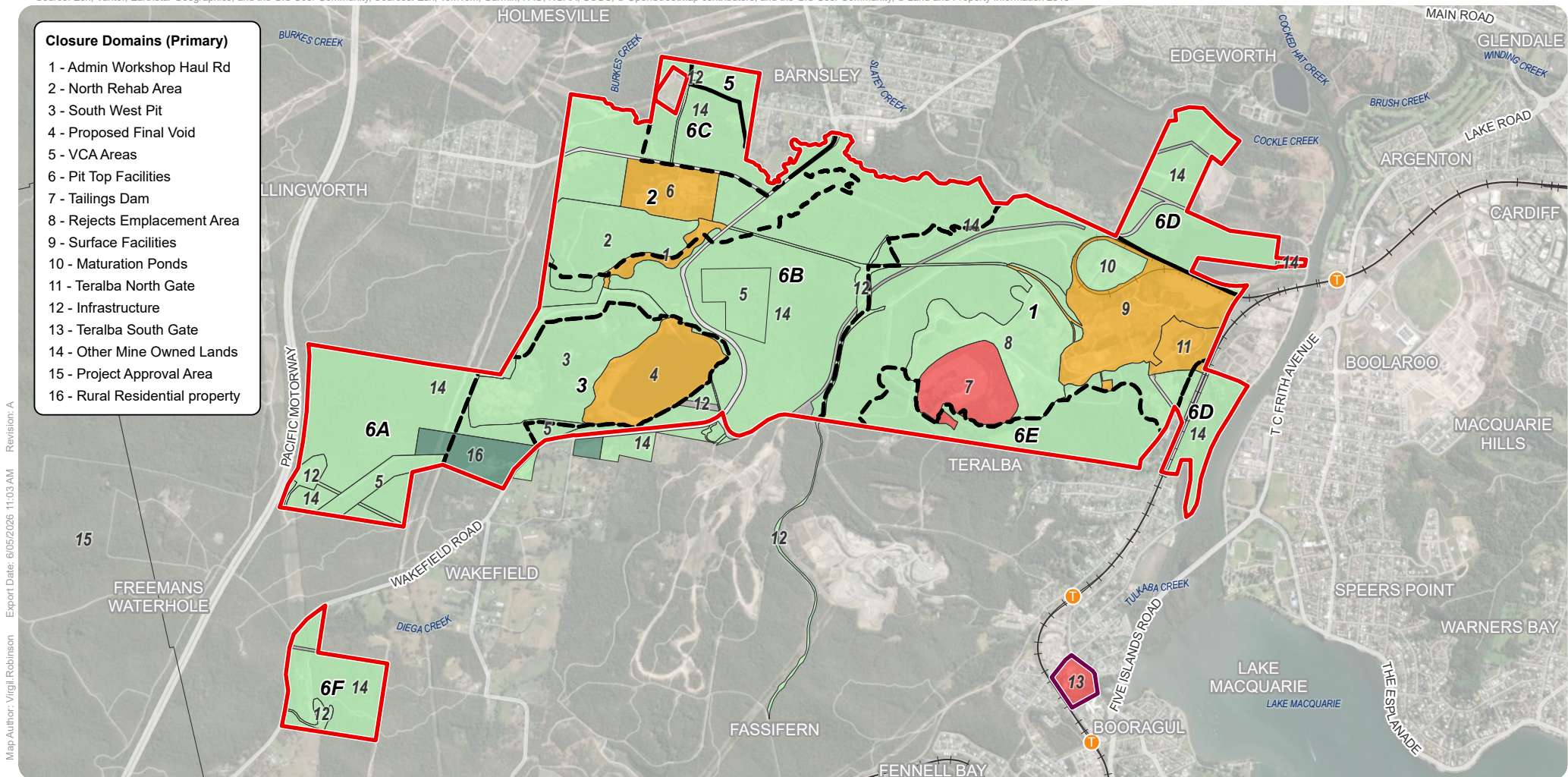


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Appendix B – Previous investigations summary

B-1. Detailed summary of investigations undertaken at each Domain

Domain 1 – Westside Administration and Workshop Area

RCA Australia (2005) — Phase 2 Environmental Site Assessment, Macquarie Coal Preparation Plant, Teralba. RCA Australia was engaged by Oceanic Coal Australia Ltd (OCAL) to undertake a Phase 2 Environmental Site Assessment across the MCPP and associated areas. Assessment criteria were based on NEPM 1999 HIL F (Commercial/Industrial). Soil sampling targeted the administration and workshop area, former underground storage tank (UST) location, oil lubricant storage area, and vehicle fleet area. No exceedances of HIL F criteria were recorded. Hydrocarbon impacts were identified in the vicinity of the former UST and oil storage area. Building footprints and the car park area were not investigated due to the presence of infrastructure.

SLR (2017) — Contaminated Site Assessment, OCAL Complex (site-wide) (Rev C, 22 February 2017). SLR undertook a site-wide CSA across Domains 1, 3, 4, 6, 9, 10, 11, 12, and 13 which included soil sampling at 250 locations, groundwater sampling at seven locations, and laboratory analysis for heavy metals, TRH, BTEXN, PAHs, phenols, OCPs, OPPs, PCBs, ASS (SPOCAS), pH, asbestos, and nutrients. No HIL/HSL (NEPM 2013) exceedances were recorded. Exceedance of the ESL criterion for TRH F2 (C10–C16) was recorded in one fill soil sample indicating limited impacts to the fill/topsoil material across this domain. Building footprints and the car park area remain largely uninvestigated.

Domain 2 – North Rehabilitation Area

No available investigation data. The domain is identified as a former rehabilitation area associated with the Westside Mine.

Domain 3 – South-West Pit / Borrow Pit Area

SLR (2017) — Contaminated Site Assessment. Soil sampling was undertaken within the borrow pit area of Domain 3. No exceedances of health or ecological investigation criteria were recorded. No available investigation data for the broader Domain 3 area outside the borrow pit.

Domain 4 – Westside Final Void / Bioremediation Facility / Sediment Disposal Area

SLR (2017) — Contaminated Site Assessment. Domain 4 was inspected as part of the pre-work assessment. No available intrusive investigation data. SLR noted that no information was available regarding the bioremediation facility or the removal of a former fuel storage element. The domain was identified as containing a proposed final void, bioremediation facility, and sediment disposal area.

Domain 5 – Voluntary Conservation Areas

No available investigation data. The domain is identified as a Voluntary Conservation Area (VCA) partially associated with the Westside Mine.

Domain 6 – West Wallsend Colliery and Pit Top Dam Areas

SLR (2017) — Contaminated Site Assessment. Domain 6 (WWC Pit Top Facilities) was assessed as part of the Precinct and Teralba Southgate CSA. Soil sampling targeted the workshop, laydown yard, coal loading facility, explosive magazine, above-ground storage tanks (ASTs), sedimentation dams, and sewage treatment facility. Exceedances of ecological criteria were recorded for heavy metals (copper, zinc) and TRH in soils. Asbestos was confirmed in soils adjacent to former infrastructure. Buried asbestos cement (AC) pipework was also identified but not fully characterised. Groundwater monitoring data was not available for Domain 6.

Coffey (2019) — Contamination Site Assessment – Catchment Dams 1, 2 and 3 (16 April 2019; Rev1, 7 August 2019). Coffey were engaged to assess three Catchment Dams (Dam 1: ~2.4 ha; Dam 2: ~1.14 ha; Dam 3: ~5 ha) located within the MCPP, which received runoff from coal processing areas, coarse rejects emplacement, product coal stockpiles, and former site infrastructure. Dam 3 has a licensed discharge point (LDP1) to Cockle Creek under EPL No. 1360. The assessment included 81 sediment samples, 6 soil samples, and 15 surface water samples, analysed for heavy metals, TRH, BTEX, PAH, PFAS, microbiological indicators, ASS (SCR/AVS), EC, pH, CEC, clay content, iron content, and TOC. Assessment criteria included both Recreational/Public Open Space and Commercial/Industrial investigation levels. Exceedances of ecological criteria were recorded for zinc in Catchment Dams 1, 2, and 4, and for chromium in Catchment Dam 4. PFAS was detected in surface water and sediments. PASS was confirmed in sediment samples. Statistical appraisal confirmed zinc exceedances above SQGs and EILs in Catchment Dam 1. No human health criteria exceedances were recorded.

Coffey (2020) — Contaminated Site Assessment – Pit Top Catchment Dams, West Wallsend Colliery. Additional assessment of the Pit Top Catchment Dams (Dams 1–4) was undertaken in 2020, building on the previous assessment completed in 2019 and included additional surface water and groundwater sampling around the Catchment Dams including upstream and downstream locations. Results confirmed residual heavy metal and PFAS impacts in sediments and surface water. Groundwater data confirmed localised impacts in the shallow unconfined aquifer.

Coffey (2019) — Sediment and Surface Water Sampling for Presence of PFAS within Dam G, Westside Mine (11 November 2019) (C105). Targeted PFAS assessment of Dam G (Westside Mine) was undertaken following identification of PFAS in the broader MCPP catchment area. Results confirmed PFAS detections in sediment and surface water within Dam G.

SLR (2015) — Hazardous Building Materials Management Survey Report. This HBM survey also included Domain 6 area as well as other domains to identify hazardous materials in structures including ACM. ACM was confirmed in multiple building components across Domain 6.

Domain 7 – Tailings Storage Facility (TSF)

No available investigation data. The TSF is understood to contain tailings from coal processing operations at the West Wallsend Colliery and Macquarie Coal Preparation Plant. A Dam Safety Review (DSR) and dam break failure impact assessment have been undertaken. The TSF is identified as a major constraint to rezoning and redevelopment of the broader Precinct.

Domain 8 – Rejects Emplacement Area

No available investigation data.

Domain 9 – MCPP Surface Facilities

RCA Australia (2005) — Phase 2 Environmental Site Assessment. RCA Australia undertook Phase 2 assessment across the MCPP surface facilities area, targeting the workshop, ASTs, former diesel UST, and surface bowser. Assessment criteria were based on NEPM 1999 HIL-F. Hydrocarbon impacts were identified in the vicinity of the former UST and diesel storage areas. No HIL-F exceedances were recorded.

SLR (2016) — Report on Intrusive Investigation of Fill Potentially Containing Furnace Slag – Coal Product Stockpile Areas (21 December 2016). SLR undertook targeted investigation of fill materials in the thermal and coking coal stockpile areas (Domain 9) to assess the presence of furnace slag. Soil sampling was undertaken at 17 borehole and test pit locations. Exceedances of Slag Investigation Criteria were recorded for arsenic, cadmium, chromium, nickel, selenium, zinc, and pH in both the thermal and coking coal stockpile areas. Results indicated that slag materials were present and exhibited geochemical characteristics warranting management as potentially reactive fill.

SLR (2017) — Contaminated Site Assessment. Domain 9 was assessed as part of the Precinct and Teralba Southgate CSA. Soil sampling targeted the MCPP surface facilities, rail loop, and coal stockpile

areas. Exceedances of ecological criteria were recorded for TRH and zinc in soils. Asbestos was confirmed near former infrastructure. Five samples were tested for ASS in Domain 9, with four sediment samples from Pond 3 and one from the rail loop. ASS testing was deemed inconclusive. Groundwater data was limited to the deeper fractured rock aquifer, with limited assessment of the shallow unconfined aquifer.

Coffey (2018) — Desktop Peer Review of Previous Site Assessment Works, West Wallsend Colliery.

Coffey undertook a desktop review of the SLR (2017) assessment, identifying data gaps which included: inadequate ASS characterisation across closure Domains 9 and 10, absence of shallow groundwater data, ecological risk uncertainties and need to compare sediment data against NSW EPA Biosolids Guidelines given treated effluent impacts.

Coffey (2019, 2020) — Catchment Dams Assessment. Catchment Dams 1–3 are located at the Domain 9/10 interface and receive runoff from Domain 9 surface facilities. Refer to Domain 6 summary above.

Domain 10 – Maturation Ponds

SLR (2017) — Contaminated Site Assessment. Domain 10 was assessed as part of the site-wide CSA. Soil and sediment sampling targeted the maturation pond areas. Exceedances of ecological criteria were recorded for heavy metals, TRH, and PAHs in maturation pond sediments. Two samples were tested for ASS in Domain 10, but no sediment samples were tested from the maturation ponds. Available groundwater data was limited.

Coffey (2019) — Maturation Pond Remediation Validation Report (Draft, 1 July 2019). Coffey undertook remediation validation of the Maturation Ponds following removal of impacted sediments. Validation sampling confirmed that remediation objectives had been achieved across most of the pond areas, with the exception of two residual locations which still returned some ecological criteria exceedances. Sediments from temporary holding cells were relocated to the Tailings Dam. ASS neutralisation was undertaken and verified. ACM was identified and removed during rehabilitation works.

Coffey (2021) — Maturation Pond Remediation Validation Report (24 February 2021). Updated validation reporting confirmed residual exceedances at two locations. Additional groundwater monitoring was recommended to assess if heavy metal and PFAS concentrations were declining following remediation. The south-eastern corner of Maturation Pond 5 (former pump/pipeline area) was not validated at the time of reporting.

Domain 11 – Teralba Northgate

RCA Australia (2005) — Phase 2 Environmental Site Assessment. RCA Australia undertook Phase 2 assessment across the Teralba Northgate area, targeting the workshop, ASTs, and former infrastructure. Hydrocarbon impacts were identified. ACM was observed in soils.

SLR (2017) — Contaminated Site Assessment. Domain 11 was assessed as part of the Precinct and Teralba Southgate CSA. Soil sampling targeted the Northgate surface facilities, mine shaft locations, and former infrastructure areas. Exceedances of ecological criteria were recorded for heavy metals and TRH. ACM was confirmed in soils, including fibrous asbestos. Methane gas was detected at mine shaft locations (Northgate Shafts 3 and 4). Various buildings and structures were inaccessible during the investigation. SLR recommended preparation of a methane management plan.

SLR (2015) — Hazardous Building Materials Management Survey. ACM confirmed in multiple building components across Domain 11.

Domain 12 – Infrastructure Corridor

SLR (2017) — Contaminated Site Assessment (as above, C101). Domain 12 was assessed as part of the Precinct and Teralba Southgate CSA. Soil sampling targeted the infrastructure corridor and haul road areas. Exceedances of ecological criteria were recorded for zinc and TRH in fill materials. Anomalous detections of

selenium and uranium were identified at a section of Haul Road. No human health criteria exceedances were recorded. PASS was identified in low-lying areas. Groundwater data was not available for Domain 12.

Domain 13 – Teralba Southgate

RCA Australia (2005) — Phase 2 Environmental Site Assessment. RCA Australia undertook Phase 2 assessment across the Teralba Southgate area, targeting the workshop, ASTs, methane plant, and surface water storage dam. Hydrocarbon impacts were identified in the vicinity of the workshop and hydrocarbon drain facility. Deep ground penetration of hydrocarbons was also observed.

SLR (2015) — Hazardous Building Materials Management Survey. ACM confirmed in multiple building components across Domain 13.

SLR (2017) — Contaminated Site Assessment. Domain 13 was assessed as part of the Precinct and Teralba Southgate CSA. Soil and sediment sampling targeted the Southgate surface facilities, methane plant area, and surface water storage dam. Exceedances of ecological criteria were recorded for heavy metals, TRH, and PAHs in soils and sediments. Deep hydrocarbon impacts were also identified. Methane gas facilities were present. Various buildings and structures were inaccessible during the investigation. Groundwater sampling was undertaken at limited locations, with potential for off-site migration of metals towards Cockle Bay identified.

Coffey (2018) — Dam Sediment Assessment, Teralba Southgate. Surface water monitoring identified seepage and overflows as potential contaminant pathways. Seven sediment and three surface water samples were analysed for COPCs against Australian and New Zealand Guidelines for Fresh and Marine Water Quality (2018) criteria. Sediment showed exceedances for TRH (C6–C36) and heavy metals (cadmium, copper, nickel, mercury, zinc) above criteria; surface water samples were below criteria. TRH concentrations were primarily attributable to coal fines with limited mobility and low subsurface leaching potential. Metal concentrations in water were low to non-detectable with minimal bioavailability and toxicity risk. The assessment concluded low risk to human health and the environment for current and proposed light industrial use; no remedial works required.

Tetra Tech Coffey (2021) — Contaminated Site Assessment, Teralba Southgate (16 November 2021). Soil sampling from 32 locations and groundwater sampling from six existing monitoring wells were undertaken to address data gaps and inform future mine closure and rehabilitation planning. No detectable methane, health-based risks, or ecological risks were identified. TRH (>C16–C34) exceedances were recorded at depth within fill materials and considered acceptable on a qualitative risk basis. Groundwater heavy metal exceedances (arsenic, copper, nickel, and zinc) were attributed to regional conditions associated with prolonged industrial activity. PFOS exceedances were identified in a down-gradient monitoring well, with detectable PFAS analytes in adjacent wells, likely associated with former firefighting activities or spills at the site. The site was considered suitable for continued commercial/industrial use, subject to appropriate maintenance. Recommendations included a hazardous ground gas assessment within enclosed structures prior to reoccupation, and three management options for PFAS: additional assessment to define source extent, containment via impervious capping, or a Human Health and Ecological Risk Assessment (HHERA) to determine tolerability of reported PFAS levels. Additional assessment would be required for any proposed change to a more sensitive land use.

Tetra Tech Coffey (2022) — Human Health and Environmental Risk Assessment, Teralba Southgate (30 August 2022). Tetra Tech Coffey undertook a Human Health and Environmental Risk Assessment (HHERA) for Domain 13. This assessment included soil, groundwater, and ground gas sampling. Exceedances of ecological criteria were confirmed for heavy metals, TRH, and PAHs. PFAS was detected in groundwater. Methane and CO₂ were detected above acceptable thresholds at mine shaft locations. The assessment confirmed that the site was not suitable for sensitive land uses without further investigation and remediation.

Domain 14 – Other Mine-Owned Lands

No available investigation data. The domain is identified as other mine-owned lands, partially associated with the Westside Mine.

Domain 15 – Project Approval Area

Domain 15 falls outside the primary physical footprint of the structure plan and zoning analysis and has been excluded from the contamination risk appraisal.

Domain 16 – Rural Residential Properties

Domain 16 comprises rural residential properties with no historical heavy industrial use. No intrusive investigation has been undertaken. The domain is expected to continue in residential use.

B-2. Summary of Acid Sulfate Soil and Salinity Assessment Data

Acid Sulfate Soils Assessment Data Summary

Site-specific ASS investigations have been undertaken across several domains, primarily focused on the MCPP sediment management areas with key findings summarised below:

- Domain 9 (MCPP Surface Facilities and Catchment Dams 1–3). Tested sediment samples (17 of 20) as part of Coffey (2019) assessment returned exceedances of the the SCR action criteria, indicating the presence of Actual Acid Sulfate Soils (AASS) and Potential Acid Sulfate Soils (PASS) within dam sediments. Samples from Dams 2 and 3 exhibited the highest sulfide concentrations and confirmed active acid generation at that time. Presence of monosulfides (AVS) was also confirmed for Dam 3. Coffey (2019) recommended preparation of an Acid Sulfate Soil Management Plan (ASSMP) prior to any dewatering or sediment disturbance. A subsequent assessment of the Pit Top Catchment Dams (Coffey, 2020) assessed four dams (Dams 1–4) at WWC. Sulfide concentrations in sediment samples were all reported below the ASS action criteria, with the exception of minor detections in Dam 1. No AVS exceedances were recorded
- Domain 10 (Maturation Ponds – MCPP). Sediments within five Maturation Ponds at the MCPP tested as part of Coffey (2018) were assessed as acid generating once dewatered. Subsequent rehabilitation works by KCE which included neutralisation with lime treatment applied to sediments from the five ponds prior to relocation to a temporary holding cell and eventual backfill into the Tailings Dam (Domain 7). Validation monitoring (Coffey, 2019 — Maturation Ponds Validation Report) recorded pH_f values of 6.0–8.1 and pH_{fox} values of 2.4–7.7 across treated stockpiles, reflecting the difficulty of achieving uniform lime mixing in heavy clay sediments. However, the risk of further acidification of treated material was assessed as low given the lined containment cell and lime application records
- Domain 12 (Infrastructure / Haul Road). SLR (2017) identified PASS in underlying natural soils based on SPOCAS analysis. An ASS management plan was recommended. ASS characterisation in this domain remains incomplete
- Domain 6 (WWC). The domain includes low-lying areas associated with former wastewater management cells (North Dam, Old Transpiration Area, Top Dam, Bottom Dam) that have not been assessed for ASS.
- Domain 14. ASS risk is mapped in low-lying areas proximate to Cockle Creek but no intrusive investigation have been undertaken at this domain to date.

SLR (2017) identified the absence of ASS assessment across the majority of the Precinct and Teralba Southgate as a data gap.

Salinity Assessment Data Summary

Site-specific salinity data across the Precinct and Teralba Southgate is limited. The available data is summarised below by domain:

- Domain 9 (MCPP Catchment Dams). Coffey (2019) collected Electrical Conductivity (EC), pH, Cation Exchange Capacity (CEC), clay content, iron content, and total organic carbon data from sediment samples within Catchment Dams 1–3 as part of the EIL derivation. Specific salinity exceedance results were not reported in the available assessment summaries. Groundwater in the vicinity of the Catchment Dams was assessed during the Coffey (2019) investigation
- Domain 10 (Maturation Ponds). The Coffey (2018) noted that sediments within the Maturation Ponds were highly saline, consistent with the operational history of the ponds receiving process water from coal preparation activities. Elevated EC was recorded in groundwater monitoring wells installed in the vicinity of the ponds. Post-rehabilitation groundwater monitoring (Coffey, 2019 — Maturation Ponds Validation Report) assessed groundwater quality following capping works. Results indicated residual groundwater quality concerns, with further monitoring recommended following completion of rehabilitation

- Domain 13 (Teralba Southgate). The Tetra Tech Coffey (2021) identified elevated strontium, iron, aluminium, and zinc in surface water samples collected from the on-site dam in 2015. These parameters are consistent with saline or mineralised groundwater discharge. Groundwater was measured at approximately 1.97 mbgs at the Teralba Southgate site, indicating shallow groundwater conditions consistent with elevated salinity risk in low-lying areas
- Domain 6 (WWC). The domain includes former wastewater management cells (North Dam, Old Transpiration Area, Top Dam, Bottom Dam) that have not been assessed for salinity or groundwater quality. Given the operational history of the WWC and the presence of mine water management infrastructure, salinity impacts to groundwater and surface water require further investigation.

No salinity-specific assessment data is available for Domains 1, 2, 3, 4, 5, 7, 8, 11, 12, 14, 15, and 16. The SLR (2017) CSA noted the absence of groundwater data across the majority of the Precinct and Teralba Southgate as a data gap, which limits the ability to characterise salinity conditions across the uninvestigated domains.

Appendix C – Summary of Data Gaps

A summary of data gaps identified at the Precinct and Teralba Southgate is provided in **Table 11-1**.

Table 11-1 Summary of Data Gaps

Domain ID	Domain name	Sub precinct IDs	Data gaps ¹
Domain 1	Westside Administration and Workshop Area	2, 6b	■ Building footprints and car park areas not investigated - inaccessible at the time of assessment ■ No groundwater data available ■ Photo-ionisation Detection (PID) screening data was not undertaken ■ PCB assessment was not undertaken ■ Hydrocarbon data previously assessed against NEPM 1999 and NSW EPA Service Station Guidelines - requires reassessment under NEPM 2013 ■ No information is available regarding the removal of a fuel storage facility from Domain 1 ■ No ASS assessment data ■ Surface water and sediment quality in Dam G and Dam J not fully assessed.
Domain 2	North Rehab Area	2, 6b	■ No intrusive investigation undertaken ■ No soil, groundwater, surface water, or sediment sampling data available ■ No ASS assessment data.
Domain 3	South-west Pit/Borrow Area	3, 6a	■ No desktop or intrusive investigation data available outside the borrow pit area ■ No information available regarding the borrow pit area used for waste storage ■ No PID screening data available ■ Hydrocarbon data previously assessed against NEPM 1999 and NSW EPA Service Station Guidelines - requires reassessment under NEPM 2013 ■ No ASS assessment data.
Domain 4	Proposed Final Void / WWC Westside Void	3	■ Limited information is available regarding the Bioremediation Facility in Domain 4 ■ Limited data was collected in peripheral areas surrounding the void ■ Sediments disposed of on north-east side of Westside Void (from Domain 6 dam clean-up) are now submerged, inaccessible and not assessed ■ No ASS assessment data.
Domain 5	Voluntary Conservation Areas	6b	■ No intrusive investigation data available for soil, groundwater, surface water or sediment ■ No ASS assessment data - ASS risk mapped but not investigated.

Domain ID	Domain name	Sub precinct IDs	Data gaps ¹
Domain 6	WWC Pit Top Facilities including Dams	2	- Water and sediment quality in wastewater cells (North Dam, Old Transpiration Area, Top Dam, Bottom Dam, Northeast Sedimentation Dam) not completely assessed - No groundwater data available to assess hydrocarbon impact on local groundwater - No measured data from methane drainage well available - Fibrous asbestos confirmed in surface soils and requires further assessment - No ASS assessment data.
Domain 7	Tailings Dam	1, 6e	- No intrusive investigation undertaken - No soil, groundwater, surface water, or sediment sampling data available - No ASS assessment data - Water and sediment quality has not been assessed. Water drains into abandoned underground workings. Further assessment needed as per <i>CLM Act</i> and the <i>POEO Act</i>.
Domain 8	Rejects Emplacement Area	1	- Limited soil, groundwater or surface water sampling data available - No ASS assessment data - No vapour/methane assessment - Slag materials not fully characterised.
Domain 9	MCPD Surface Facilities	1, 6d	- Further characterisation of contamination required - PID screening data not undertaken - Further ASS characterisation needed - Extent of asbestos pipework needs to be further defined.
Domain 10	Maturation Ponds	1, 6d	- Sediment characterisation incomplete - Water and sediment quality in Maturation Ponds 1–5 not assessed - Limited groundwater data - Limited ASS characterisation completed - Buried asbestos pipework presence confirmed but extent unknown.
Domain 11	Teralba North Gate	1, 6d	- Building footprints and inaccessible structures not investigated - No groundwater data collected - No surface water assessment - Detailed methane gas assessment not undertaken - No ASS assessment.

Domain ID	Domain name	Sub precinct IDs	Data gaps ¹
Domain 12	Infrastructure (Haul Road)	1, 2, 5, 6a, 6b, 6c, 6f	- Anomalous selenium and uranium enrichment identified but not further assessed - No groundwater data - ASS identified in underlying natural soils but not fully characterised - No surface water assessment.
Domain 13	Teralba South Gate	Not applicable	- Further assessment of soil and groundwater is required - Sampling beneath workshop has not been completed - Asbestos delineation sampling not undertaken - Surface water sampling has not been completed - Detailed ground gas assessments have not been undertaken.
Domain 14	Other Mine Owned Lands	1, 2, 3, 5, 6a, 6b, 6c, 6d, 6e, 6f	- No intrusive investigation undertaken - No soil, groundwater, or surface water data - ASS risk mapped but not field-verified.
Domain 15	Rural Residential Property	3, 6a, 6b	No intrusive investigation undertaken.
Domain 16	Westside Administration and Workshop Area	3	No intrusive investigation undertaken.

Table notes:

¹ Based on the data and information provided in previous investigation reports (Section 3.3)

Appendix D – Domain risk qualifications

Table 11-2 presents the application of the domain constraint qualification to domains included in the CSM (**Section 4**).

Table 11-2 Domain constraint rating qualifications

Domain ID	Domain Name	Consequence Rating	Likelihood rating	Constraint Rating
1	Westside Administration and Workshop Area	Moderate	Possible	Moderate
2	North Rehabilitation Area	Mild	Possible	Low
3	Borrow Pit Area	Mild	Possible	Low
4	Westside Final Void / WWC Westside Void	Moderate	Possible	Moderate
5	Voluntary Conservation Areas	Mild	Unlikely	Low
6	WWC Pit Top Facilities	Moderate	Possible	Moderate
7	Tailings Storage Facility	High	Possible	Moderate to high
8	MCCP Rejects Emplacement Area	Mild	Possible	Low
9	Washery and Workshop Area	Moderate	Possible	Moderate
10	Maturation Ponds	Mild	Unlikely	Low
11	Teralba Northgate	Moderate	Possible	Moderate
12	Infrastructure (Haul Road)	Mild	Unlikely	Low
13	Teralba Southgate	Moderate	Likely	Moderate to high
14	Other Mine Owned Lands	Mild	Unlikely	Low
16	Rural Residential Property	Negligible	Unlikely	Negligible

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